

Earlier T took the derivative and S took the integral. The transformation TS is the identity but not ST . Therefore AB is the identity matrix but not BA :

$$AB = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} = I \quad \text{but} \quad BA = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The Identity Transformation and the Change of Basis Matrix

We now find the matrix for the special and boring transformation $T(v) = v$. This **identity transformation** does nothing to v . The matrix for $T = I$ also does nothing, *provided* the output basis is the same as the input basis. The output $T(v_1)$ is v_1 . When the bases are the same, this is w_1 . So the first column of A is $(1, 0, \dots, 0)$.

When each output $T(v_j) = v_j$ is the same as w_j , the matrix is just I .

This seems reasonable: The identity transformation is represented by the identity matrix. But suppose the bases are *different*. Then $T(v_1) = v_1$ is a combination of the w 's. That combination $m_{11}w_1 + \dots + m_{n1}w_n$ tells the first column of the matrix (call it M).

Identity transformation

When the outputs $T(v_j) = v_j$ are combinations $\sum_{i=1}^n m_{ij}w_i$, the "change of basis matrix" is M .

The basis is changing but the vectors themselves are not changing: $T(v) = v$. When the inputs have one basis and the outputs have another basis, the matrix is not I .

Example 9 The input basis is $v_1 = (3, 7)$ and $v_2 = (2, 5)$. The output basis is $w_1 = (1, 0)$ and $w_2 = (0, 1)$. Then the matrix M is easy to compute:

Change of basis The matrix for $T(v) = v$ is $M = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$.

Reason The first input is the basis vector $v_1 = (3, 7)$. The output is also $(3, 7)$ which we express as $3w_1 + 7w_2$. Then the first column of M contains 3 and 7.

This seems too simple to be important. It becomes trickier when the change of basis goes the other way. We get the inverse of the previous matrix M :

Example 10 The input basis is now $v_1 = (1, 0)$ and $v_2 = (0, 1)$. The outputs are just $T(v) = v$. But the output *basis* is now $w_1 = (3, 7)$ and $w_2 = (2, 5)$.

**Reverse the bases
Invert the matrix**

The matrix for $T(v) = v$ is $\begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}^{-1} = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$.

Reason The first input is $v_1 = (1, 0)$. The output is also v_1 but we express it as $5w_1 - 7w_2$. Check that $5(3, 7) - 7(2, 5)$ does produce $(1, 0)$. We are combining the columns of the previous M to get the columns of I . The matrix to do that is M^{-1} .

$$\begin{array}{l} \text{Change basis} \\ \text{Change back} \end{array} \quad [w_1 \ w_2] \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix} = [v_1 \ v_2] \text{ is } MM^{-1} = I.$$

A mathematician would say that the matrix MM^{-1} corresponds to the product of two identity transformations. We start and end with the same basis $(1, 0)$ and $(0, 1)$. Matrix multiplication must give I . So the two change of basis matrices are inverses.

One thing is sure. Multiplying A times $(1, 0, \dots, 0)$ gives column 1 of the matrix. The novelty of this section is that $(1, 0, \dots, 0)$ stands for the first vector v_1 , *written in the basis of v 's*. Then column 1 of the matrix is that same vector v_1 , *written in the standard basis*.

Wavelet Transform = Change to Wavelet Basis

Wavelets are little waves. They have different lengths and they are localized at different places. The first basis vector is not actually a wavelet, it is the very useful flat vector of all ones. This example shows “Haar wavelets”:

$$\text{Haar basis} \quad w_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad w_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} \quad w_3 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \quad w_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}. \quad (8)$$

Those vectors are *orthogonal*, which is good. You see how w_3 is localized in the first half and w_4 is localized in the second half. The **wavelet transform** finds the coefficients c_1, c_2, c_3, c_4 when the input signal $v = (v_1, v_2, v_3, v_4)$ is expressed in the wavelet basis:

$$\text{Transform } v \text{ to } c \quad v = c_1 w_1 + c_2 w_2 + c_3 w_3 + c_4 w_4 = Wc \quad (9)$$

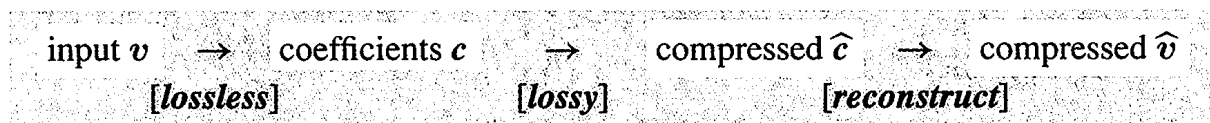
The coefficients c_3 and c_4 tell us about details in the first half and last half of v . The coefficient c_1 is the average.

Why do want to change the basis? I think of v_1, v_2, v_3, v_4 as the intensities of a signal. In audio they are volumes of sound. In images they are pixel values on a scale of black to white. An electrocardiogram is a medical signal. Of course $n = 4$ is very short, and $n = 10,000$ is more realistic. We may need to *compress* that long signal, by keeping only the largest 5% of the coefficients. This is 20 : 1 compression and (to give only two of its applications) it makes High Definition TV and video conferencing possible.

If we keep only 5% of the *standard* basis coefficients, we lose 95% of the signal. In image processing, 95% of the image disappears. In audio, 95% of the tape goes blank. But if we choose a better basis of w 's, 5% of the basis vectors can combine to come very close to the original signal. In image processing and audio coding, you can't see or hear the difference. We don't need the other 95%!

One good basis vector is the flat $(1, 1, 1, 1)$. That part alone can represent the constant background of our image. A short wave like $(0, 0, 1, -1)$ or in higher dimensions $(0, 0, 0, 0, 0, 0, 1, -1)$ represents a detail at the end of the signal.

The three steps are the transform and compression and inverse transform.



In linear algebra, where everything is perfect, we omit the compression step. The output $\hat{v}$ is exactly the same as the input v . The transform gives $c = W^{-1}v$ and the reconstruction brings back $v = Wc$. In true signal processing, where nothing is perfect but everything is fast, the transform (lossless) and the compression (which only loses unnecessary information) are absolutely the keys to success. The output is $\hat{v} = W\hat{c}$.

I will show those steps for a typical vector like $v = (6, 4, 5, 1)$. Its wavelet coefficients are $c = (4, 1, 1, 2)$. The reconstruction $4w_1 + w_2 + w_3 + 2w_4$ is $v = Wc$:

$$\begin{bmatrix} 6 \\ 4 \\ 5 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 1 \\ 2 \end{bmatrix}. \quad (10)$$

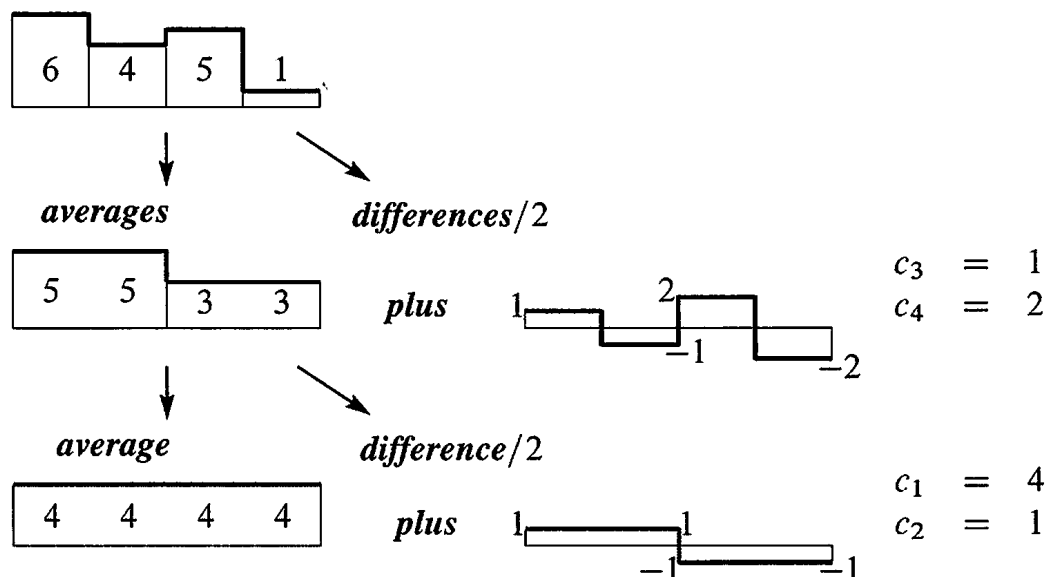
Those coefficients c are $W^{-1}v$. Inverting this basis matrix W is easy because the w 's in its columns are orthogonal. But they are not unit vectors, so rescale:

$$W^{-1} = \begin{bmatrix} \frac{1}{4} & & & \\ & \frac{1}{4} & & \\ & & \frac{1}{2} & \\ & & & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}.$$

The $\frac{1}{4}$'s in the first row of $c = W^{-1}v$ mean that $c_1 = 4$ is the average of 6, 4, 5, 1.

Example 11 (Same wavelet basis by recursion) I can't resist showing you a faster way to find the c 's. The special point of the wavelet basis is that you can pick off the details in c_3 and c_4 , before the coarse details in c_2 and the overall average in c_1 . A picture will explain this "multiscale" method, which is in Chapter 1 of my textbook with Nguyen on *Wavelets and Filter Banks* (Wellesley-Cambridge Press).

Split $v = (6, 4, 5, 1)$ into averages and waves at small scale and then large scale:



Fourier Transform (DFT) = Change to Fourier Basis

The first thing an electrical engineer does with a signal is to take its Fourier transform. For finite vectors we are speaking about the **Discrete Fourier Transform**. The DFT involves complex numbers (powers of $e^{2\pi i/n}$). But if we choose $n = 4$, the matrices are small and the only complex numbers are i and $i^3 = -i$. A true electrical engineer would write j instead of i for $\sqrt{-1}$.

$$\begin{array}{l} \text{Fourier basis } w_1 \text{ to } w_n \\ \text{in the columns of } F \end{array} \quad F = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix}$$

The first column is the useful flat basis vector $(1, 1, 1, 1)$. It represents the average signal or the direct current (the DC term). It is a wave at zero frequency. The third column is $(1, -1, 1, -1)$, which alternates at the highest frequency. *The Fourier transform decomposes the signal into waves at equally spaced frequencies.*

The Fourier matrix F is absolutely the most important complex matrix in mathematics and science and engineering. Section 10.3 of this book explains the **Fast Fourier Transform**: it can be seen as a factorization of F into matrices with many zeros. The FFT has revolutionized entire industries, by speeding up the Fourier transform. The beautiful thing is that F^{-1} looks like F , with i changed to $-i$:

$$\begin{array}{l} \text{Fourier transform } v \text{ to } c \\ v = c_1 w_1 + \cdots + c_n w_n = Fc \\ \text{Fourier coefficients } c = F^{-1}v \end{array} \quad F^{-1} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & (-i) & (-i)^2 & (-i)^3 \\ 1 & (-i)^2 & (-i)^4 & (-i)^6 \\ 1 & (-i)^3 & (-i)^6 & (-i)^9 \end{bmatrix} = \frac{1}{4} \overline{F}.$$

The MATLAB command $c = \text{fft}(v)$ produces the Fourier coefficients $c_1, \dots, c_n$ of the vector v . It multiplies v by F^{-1} (fast).

■ REVIEW OF THE KEY IDEAS ■

1. If we know $T(v_1), \dots, T(v_n)$ for a basis, linearity will determine all other $T(v)$.
2. $\left\{ \begin{array}{l} \text{Linear transformation } T \\ \text{Input basis } v_1, \dots, v_n \\ \text{Output basis } w_1, \dots, w_m \end{array} \right\} \rightarrow \begin{array}{l} \text{Matrix } A \text{ (} m \text{ by } n \text{)} \\ \text{represents } T \\ \text{in these bases} \end{array}$
3. The derivative and integral matrices are one-sided inverses: $d(\text{constant})/dx = 0$:

(Derivative) (Integral) = I is the Fundamental Theorem of Calculus.

4. If A and B represent T and S , and the output basis for S is the input basis for T , then the matrix AB represents the transformation $T(S(u))$.

5. The change of basis matrix M represents $T(v) = v$. Its columns are the coefficients of the output basis expressed in the input basis: $w_j = m_{1j}v_1 + \cdots + m_{nj}v_n$.

■ WORKED EXAMPLES ■

7.2 A Using the standard basis, find the 4 by 4 matrix P that represents a *cyclic permutation* T from $x = (x_1, x_2, x_3, x_4)$ to $T(x) = (x_4, x_1, x_2, x_3)$. Find the matrix for T^2 . What is the triple shift $T^3(x)$ and why is $T^3 = T^{-1}$?

Find two real independent eigenvectors of P . What are all the eigenvalues of P ?

Solution The first vector $(1, 0, 0, 0)$ in the standard basis transforms to $(0, 1, 0, 0)$ which is the second basis vector. So the first column of P is $(0, 1, 0, 0)$. The other three columns come from transforming the other three standard basis vectors:

$$P = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad \text{Then } P \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_4 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ copies } T.$$

Since we used the standard basis, T is ordinary multiplication by P . The matrix for T^2 is a “double cyclic shift” P^2 and it produces (x_3, x_4, x_1, x_2) .

The triple shift T^3 will transform $x = (x_1, x_2, x_3, x_4)$ to $T^3(x) = (x_2, x_3, x_4, x_1)$. If we apply T once more we are back to the original x . So $T^4 = \text{identity transformation}$ and $P^4 = \text{identity matrix}$.

Two real eigenvectors of P are $(1, 1, 1, 1)$ with eigenvalue $\lambda = 1$ and $(1, -1, 1, -1)$ with eigenvalue $\lambda = -1$. The shift leaves $(1, 1, 1, 1)$ unchanged and it reverses signs in $(1, -1, 1, -1)$. The other eigenvalues are i and $-i$. The determinant is $\lambda_1\lambda_2\lambda_3\lambda_4 = -1$.

Notice that the eigenvalues $1, i, -1, -i$ add to zero (the trace of P). They are the 4th roots of 1, since $\det(P - \lambda I) = \lambda^4 - 1$. They are at angles $0^\circ, 90^\circ, 180^\circ, 270^\circ$ in the complex plane. **The Fourier matrix F is the eigenvector matrix for P .**

7.2 B The space of 2 by 2 matrices has these four “vectors” as a basis:

$$v_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad v_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

T is the linear transformation that *transposes* every 2 by 2 matrix. What is the matrix A that represents T in this basis (output basis = input basis)? What is the inverse matrix A^{-1} ? What is the transformation T^{-1} that inverts the transpose operation?

Solution Transposing those four “basis matrices” just reverses v_2 and v_3 :

$$\begin{array}{l} T(v_1) = v_1 \\ T(v_2) = v_3 \\ T(v_3) = v_2 \\ T(v_4) = v_4 \end{array} \quad \text{gives the four columns of} \quad A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The inverse matrix A^{-1} is the same as A . The inverse transformation T^{-1} is the same as T . If we transpose and transpose again, the final output equals the original input.

Problem Set 7.2

Questions 1–4 extend the first derivative example to higher derivatives.

- 1 The transformation S takes the *second derivative*. Keep $1, x, x^2, x^3$ as the basis v_1, v_2, v_3, v_4 and also as w_1, w_2, w_3, w_4 . Write Sw_1, Sw_2, Sw_3, Sw_4 in terms of the w 's. Find the 4 by 4 matrix B for S .
- 2 What functions have $v'' = 0$? They are in the kernel of the second derivative S . What vectors are in the nullspace of its matrix B in Problem 1?
- 3 B is not the square of a rectangular first derivative matrix:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \text{ does not allow } A^2.$$

Add a zero row to A , so that output space = input space. Compare A^2 with B . Conclusion: For $B = A^2$ we want output basis = _____ basis. Then $m = n$.

- 4 (a) The product TS of first and second derivatives produces the *third derivative*. Add zeros to make 4 by 4 matrices, then compute AB .
(b) The matrix B^2 corresponds to $S^2 = \text{fourth derivative}$. Why is this zero?

Questions 5–9 are about a particular T and its matrix A .

- 5 With bases v_1, v_2, v_3 and w_1, w_2, w_3 , suppose $T(v_1) = w_2$ and $T(v_2) = T(v_3) = w_1 + w_3$. T is a linear transformation. Find the matrix A and multiply by the vector $(1, 1, 1)$. What is the output from T when the input is $v_1 + v_2 + v_3$?
- 6 Since $T(v_2) = T(v_3)$, the solutions to $T(v) = 0$ are $v = \underline{\hspace{2cm}}$. What vectors are in the nullspace of A ? Find all solutions to $T(v) = w_2$.
- 7 Find a vector that is not in the column space of A . Find a combination of w 's that is not in the range of T .
- 8 You don't have enough information to determine T^2 . Why is its matrix not necessarily A^2 ? What more information do you need?

- 9 Find the rank of A . This is not the dimension of the output space $\mathbf{W}$. It is the dimension of the _____ of T .

Questions 10–13 are about invertible linear transformations.

- 10 Suppose $T(v_1) = w_1 + w_2 + w_3$ and $T(v_2) = w_2 + w_3$ and $T(v_3) = w_3$. Find the matrix A for T using these basis vectors. What input vector v gives $T(v) = w_1$?
- 11 Invert the matrix A in Problem 10. Also invert the transformation T —what are $T^{-1}(w_1)$ and $T^{-1}(w_2)$ and $T^{-1}(w_3)$?
- 12 Which of these are true and why is the other one ridiculous?
- (a) $T^{-1}T = I$ (b) $T^{-1}(T(v_1)) = v_1$ (c) $T^{-1}(T(w_1)) = w_1$.
- 13 Suppose the spaces $\mathbf{V}$ and $\mathbf{W}$ have the same basis v_1, v_2 .
- (a) Describe a transformation T (not I) that is its own inverse.
- (b) Describe a transformation T (not I) that equals T^2 .
- (c) Why can't the same T be used for both (a) and (b)?

Questions 14–19 are about changing the basis.

- 14 (a) What matrix transforms $(1, 0)$ into $(2, 5)$ and transforms $(0, 1)$ to $(1, 3)$?
- (b) What matrix transforms $(2, 5)$ to $(1, 0)$ and $(1, 3)$ to $(0, 1)$?
- (c) Why does no matrix transform $(2, 6)$ to $(1, 0)$ and $(1, 3)$ to $(0, 1)$?
- 15 (a) What matrix M transforms $(1, 0)$ and $(0, 1)$ to (r, t) and (s, u) ?
- (b) What matrix N transforms (a, c) and (b, d) to $(1, 0)$ and $(0, 1)$?
- (c) What condition on a, b, c, d will make part (b) impossible?
- 16 (a) How do M and N in Problem 15 yield the matrix that transforms (a, c) to (r, t) and (b, d) to (s, u) ?
- (b) What matrix transforms $(2, 5)$ to $(1, 1)$ and $(1, 3)$ to $(0, 2)$?
- 17 If you keep the same basis vectors but put them in a different order, the change of basis matrix M is a _____ matrix. If you keep the basis vectors in order but change their lengths, M is a _____ matrix.
- 18 The matrix that rotates the axis vectors $(1, 0)$ and $(0, 1)$ through an angle θ is Q . What are the coordinates (a, b) of the original $(1, 0)$ using the new (rotated) axes? This *inverse* can be tricky. Draw a figure or solve for a and b :

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} = a \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} + b \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}.$$

- 19** The matrix that transforms $(1, 0)$ and $(0, 1)$ to $(1, 4)$ and $(1, 5)$ is $M = \underline{\hspace{2cm}}$. The combination $a(1, 4) + b(1, 5)$ that equals $(1, 0)$ has $(a, b) = (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$. How are those new coordinates of $(1, 0)$ related to M or M^{-1} ?

Questions 20–23 are about the space of quadratic polynomials $A + Bx + Cx^2$.

- 20** The parabola $w_1 = \frac{1}{2}(x^2 + x)$ equals one at $x = 1$, and zero at $x = 0$ and $x = -1$. Find the parabolas w_2, w_3 , and then find $y(x)$ by linearity.

- (a) w_2 equals one at $x = 0$ and zero at $x = 1$ and $x = -1$.
 (b) w_3 equals one at $x = -1$ and zero at $x = 0$ and $x = 1$.
 (c) $y(x)$ equals 4 at $x = 1$ and 5 at $x = 0$ and 6 at $x = -1$. Use w_1, w_2, w_3 .

- 21** One basis for second-degree polynomials is $v_1 = 1$ and $v_2 = x$ and $v_3 = x^2$. Another basis is w_1, w_2, w_3 from Problem 20. Find two change of basis matrices, from the w 's to the v 's and from the v 's to the w 's.

- 22** What are the three equations for A, B, C if the parabola $Y = A + Bx + Cx^2$ equals 4 at $x = a$ and 5 at $x = b$ and 6 at $x = c$? Find the determinant of the 3 by 3 matrix. That matrix transforms values like 4, 5, 6 to parabolas—or is it the other way?

- 23** Under what condition on the numbers $m_1, m_2, \dots, m_9$ do these three parabolas give a basis for the space of all parabolas?

$$v_1 = m_1 + m_2x + m_3x^2, \quad v_2 = m_4 + m_5x + m_6x^2, \quad v_3 = m_7 + m_8x + m_9x^2.$$

- 24** The Gram-Schmidt process changes a basis a_1, a_2, a_3 to an orthonormal basis q_1, q_2, q_3 . These are columns in $A = QR$. Show that R is the change of basis matrix from the a 's to the q 's (a_2 is what combination of q 's when $A = QR$?).

- 25** Elimination changes the rows of A to the rows of U with $A = LU$. Row 2 of A is what combination of the rows of U ? Writing $A^T = U^T L^T$ to work with columns, the change of basis matrix is $M = L^T$. (We have bases provided the matrices are $\underline{\hspace{2cm}}$.)

- 26** Suppose v_1, v_2, v_3 are **eigenvectors** for T . This means $T(v_i) = \lambda_i v_i$ for $i = 1, 2, 3$. What is the matrix for T when the input and output bases are the v 's?

- 27** Every invertible linear transformation can have I as its matrix! Choose any input basis $v_1, \dots, v_n$. For output basis choose $w_i = T(v_i)$. Why must T be invertible?

- 28** Using $v_1 = w_1$ and $v_2 = w_2$ find the standard matrix for these T 's:

$$(a) \quad T(v_1) = \mathbf{0} \text{ and } T(v_2) = 3v_1 \qquad (b) \quad T(v_1) = v_1 \text{ and } T(v_1 + v_2) = v_1.$$

- 29** Suppose T is reflection across the x axis and S is reflection across the y axis. The domain V is the xy plane. If $v = (x, y)$ what is $S(T(v))$? Find a simpler description of the product ST .

- 30** Suppose T is reflection across the 45° line, and S is reflection across the y axis. If $v = (2, 1)$ then $T(v) = (1, 2)$. Find $S(T(v))$ and $T(S(v))$. This shows that generally $ST \neq TS$.
- 31** Show that the product ST of two reflections is a rotation. Multiply these reflection matrices to find the rotation angle:

$$\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}.$$

- 32** True or false: If we know $T(v)$ for n different nonzero vectors in $\mathbf{R}^n$, then we know $T(v)$ for every vector in $\mathbf{R}^n$.
- 33** Express $e = (1, 0, 0, 0)$ and $v = (1, -1, 1, -1)$ in the wavelet basis, as in equations (8-10). The coefficients c_1, c_2, c_3, c_4 solve $Wc = e$ and $Wc = v$.
- 34** To represent $v = (7, 5, 3, 1)$ in the wavelet basis, start with $(6, 6, 2, 2) + (1, -1, 1, -1)$. Then write $6, 6, 2, 2$ as an overall average plus a difference, using $1, 1, 1, 1$ and $1, 1, -1, -1$.
- 35** What are the eight vectors in the wavelet basis for $\mathbf{R}^8$? They include the long wavelet $(1, 1, 1, 1, -1, -1, -1, -1)$ and the short wavelet $(1, -1, 0, 0, 0, 0, 0, 0)$.
- 36** Suppose we have two bases $v_1, \dots, v_n$ and $w_1, \dots, w_n$ for $\mathbf{R}^n$. If a vector has coefficients b_i in one basis and c_i in the other basis, what is the change of basis matrix in $b = Mc$? Start from

$$b_1 v_1 + \dots + b_n v_n = Vb = c_1 w_1 + \dots + c_n w_n = Wc.$$

Your answer represents $T(v) = v$ with input basis of v 's and output basis of w 's. Because of different bases, the matrix is not I .

Challenge Problems

- 37** The space $\mathbf{M}$ of 2 by 2 matrices has the basis v_1, v_2, v_3, v_4 in Worked Example 7.2 B. Suppose T multiplies each matrix by $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$. What 4 by 4 matrix A represents this transformation T on matrix space?
- 38** Suppose A is a 3 by 4 matrix of rank $r = 2$, and $T(v) = Av$. Choose input basis vectors v_1, v_2 from the row space of A and v_3, v_4 from the nullspace. Choose output basis $w_1 = Av_1, w_2 = Av_2$ in the column space and w_3 from the nullspace of A^T . What specially simple matrix represents this T in these special bases?

7.3 Diagonalization and the Pseudoinverse

This section produces better matrices by choosing better bases. When the goal is a diagonal matrix, one way is a basis of *eigenvectors*. The other way is two bases (the input and output bases are different). Those left and right *singular vectors* are orthonormal basis vectors for the four fundamental subspaces of A . They come from the SVD.

By reversing those input and output bases, we will find the “pseudoinverse” of A . This matrix A^+ sends $\mathbf{R}^m$ back to $\mathbf{R}^n$, and it sends column space back to row space.

The truth is that all our great factorizations of A can be regarded as a change of basis. But this is a short section, so we concentrate on the two outstanding examples. In both cases the good matrix is *diagonal*. It is Λ with one basis or Σ with two bases.

1. $S^{-1}AS = \Lambda$ when the input and output bases are eigenvectors of A .
2. $U^{-1}AV = \Sigma$ when those bases are eigenvectors of $A^T A$ and AA^T .

You see immediately the difference between Λ and Σ . In Λ the bases are the same. Then $m = n$ and the matrix A must be square. And some square matrices cannot be diagonalized by any S , because they don't have n independent eigenvectors.

In Σ the input and output bases are different. The matrix A can be rectangular. The bases are *orthonormal* because $A^T A$ and AA^T are symmetric. Then $U^{-1} = U^T$ and $V^{-1} = V^T$. Every matrix A is allowed, and A has the diagonal form Σ . This is the Singular Value Decomposition (SVD) of Section 6.7.

The eigenvector basis is orthonormal only when $A^T A = AA^T$ (a “normal” matrix). That includes symmetric and antisymmetric and orthogonal matrices (*special* might be a better word than normal). In this case the singular values in Σ are the absolute values $\sigma_i = |\lambda_i|$, so that $\Sigma = \mathbf{abs}(\Lambda)$. The two diagonalizations are the same when $A^T A = AA^T$, except for possible factors -1 (real) and $e^{i\theta}$ (complex).

I will just note that the Gram-Schmidt factorization $A = QR$ chooses only *one* new basis. That is the orthogonal output basis given by Q . The input uses the standard basis given by I . We don't reach a diagonal Σ , but we do reach a triangular R . The output basis matrix appears on the left and the input basis appears on the right, in $A = QRI$.

We start with input basis equal to output basis. That will produce S and S^{-1} .

Similar Matrices: A and $S^{-1}AS$ and $W^{-1}AW$

Begin with a square matrix and one basis. The input space $\mathbf{V}$ is $\mathbf{R}^n$ and the output space $\mathbf{W}$ is also $\mathbf{R}^n$. The standard basis vectors are the columns of I . The matrix is n by n , and we call it A . The linear transformation T is “multiplication by A ”.

Most of this book has been about one fundamental problem—to make the matrix simple. We made it triangular in Chapter 2 (by elimination) and Chapter 4 (by Gram-Schmidt). We made it diagonal in Chapter 6 (by eigenvectors). Now that change from A to Λ comes from a *change of basis: Eigenvalue matrix from eigenvector basis*.

Here are the main facts in advance. When you change the basis for V , the matrix changes from A to AM . Because V is the input space, the matrix M goes on the right (to come first). When you change the basis for W , the new matrix is $M^{-1}A$. We are working with the output space so M^{-1} is on the left (to come last).

If you change both bases in the same way, the new matrix is $M^{-1}AM$. The good basis vectors are the eigenvectors of A , when the matrix becomes $S^{-1}AS = \Lambda$.

When the basis contains the eigenvectors $x_1, \dots, x_n$, the matrix for T becomes Λ .

Reason To find column 1 of the matrix, input the first basis vector x_1 . The transformation multiplies by A . The output is $Ax_1 = \lambda_1 x_1$. This is λ_1 times the first basis vector plus zero times the other basis vectors. Therefore the first column of the matrix is $(\lambda_1, 0, \dots, 0)$.

In the eigenvector basis, the matrix is diagonal.

Example 1 Project onto the line $y = -x$ that goes from northwest to southeast. The vector $(1, 0)$ projects to $(.5, -.5)$ on that line. The projection of $(0, 1)$ is $(-.5, .5)$:

1. **Standard matrix: Project standard basis**
$$A = \begin{bmatrix} .5 & -.5 \\ -.5 & .5 \end{bmatrix}.$$

2. **Find the diagonal matrix Λ in the eigenvector basis.**

Solution The eigenvectors for this projection are $x_1 = (1, -1)$ and $x_2 = (1, 1)$. The first eigenvector lies on the 135° line and the second is perpendicular (on the 45° line). Their projections are x_1 and 0 . The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 0$.

2. **Diagonalized matrix: Project eigenvectors**
$$\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

3. **Find a third matrix B using another basis $v_1 = w_1 = (2, 0)$ and $v_2 = w_2 = (1, 1)$.**

Solution w_1 is not an eigenvector, so the matrix B in this basis will not be diagonal. The first way to compute B follows the rule of Section 7.2:

Find column j of the matrix by writing the projection $T(v_j)$ as a combination of w 's.

Apply the projection T to $(2, 0)$. The result is $(1, -1)$ which is $w_1 - w_2$. So the first column of B contains 1 and -1 . The second vector $w_2 = (1, 1)$ projects to zero, so the second column of B contains 0 and 0. The eigenvalues must stay at 1 and 0:

3. **Third similar matrix: Project w_1 and w_2**
$$B = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}. \quad (1)$$

The second way to find the same B is more insightful. Use W^{-1} and W to change between the standard basis and the basis of w 's. Those change of basis matrices are

representing the identity transformation! The product of transformations is just ITI . The product of matrices is $B = W^{-1}AW$. This approach shows that B is *similar* to A .

For any basis $w_1, \dots, w_n$ find the matrix B in three steps. Change the input basis to the standard basis with W . The matrix in the standard basis is A . Change the output basis back to the w 's with W^{-1} . Then $B = W^{-1}AW$ represents ITI :

$$B_{w's \text{ to } w's} = W_{\text{standard to } w's}^{-1} A_{\text{standard}} W_{w's \text{ to standard}} \quad (2)$$

A change of basis produces a similarity transformation to $W^{-1}AW$ in the matrix.

Example 2 (continuing with the projection) Apply this $W^{-1}AW$ rule to find B , when the basis $(2, 0)$ and $(1, 1)$ is in the columns of W :

$$W^{-1}AW = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}.$$

The $W^{-1}AW$ rule has produced the same B as in equation (1). The matrices A and B are similar. They have the same eigenvalues (1 and 0). And Λ is similar too.

Notice that the projection matrix keeps the property $A^2 = A$ and $B^2 = B$ and $\Lambda^2 = \Lambda$. The second projection doesn't move the first projection.

The Singular Value Decomposition (SVD)

Now the input basis $v_1, \dots, v_n$ can be different from the output basis $u_1, \dots, u_m$. In fact the input space $\mathbf{R}^n$ can be different from the output space $\mathbf{R}^m$. Again the best matrix is diagonal (now m by n). To achieve this diagonal matrix Σ , each input vector v_j must transform into a multiple of the output vector u_j . That multiple is the *singular value* σ_j on the main diagonal of Σ :

$$\text{SVD} \quad Av_j = \begin{cases} \sigma_j u_j & \text{for } j \leq r \\ \mathbf{0} & \text{for } j > r \end{cases} \quad \text{with orthonormal bases.} \quad (3)$$

The singular values are in the order $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$. The rank r enters because (by definition) singular values are not zero. The second part of the equation says that v_j is in the nullspace for $j = r + 1, \dots, n$. This gives the correct number $n - r$ of basis vectors for the nullspace.

Let me connect the matrices with the linear transformations they represent. A and Σ represent the *same transformation*. $A = U\Sigma V^T$ uses the standard bases for $\mathbf{R}^n$ and $\mathbf{R}^m$. The diagonal Σ uses the input basis of v 's and the output basis of u 's. The orthogonal matrices V and U give the basis changes; they represent the identity transformations (in $\mathbf{R}^n$ and $\mathbf{R}^m$). The product of transformations is ITI , and it is represented in the v and u bases by $U^{-1}AV$ which is Σ .

The matrix Σ in the u and v bases comes from A in the standard bases by $U^{-1}AV$:

$$\Sigma_{v's \text{ to } u's} = U_{\text{standard to } u's}^{-1} A_{\text{standard}} V_{v's \text{ to standard}}. \quad (4)$$

The SVD chooses orthonormal bases ($U^{-1} = U^T$ and $V^{-1} = V^T$) that diagonalize A .

The two orthonormal bases in the SVD are the eigenvector bases for $A^T A$ (the v 's) and AA^T (the u 's). Since those are symmetric matrices, their unit eigenvectors are orthonormal. Their eigenvalues are the numbers σ_j^2 . Equations (10) and (11) in Section 6.7 proved that those bases diagonalize the standard matrix A to produce Σ .

Polar Decomposition

Every complex number has the polar form $re^{i\theta}$. A nonnegative number r multiplies a number on the unit circle. (Remember that $|e^{i\theta}| = |\cos \theta + i \sin \theta| = 1$.) Thinking of these numbers as 1 by 1 matrices, $r \geq 0$ corresponds to a *positive semidefinite matrix* (call it H) and $e^{i\theta}$ corresponds to an *orthogonal matrix* Q . The **polar decomposition** extends this factorization to matrices: orthogonal times semidefinite, $A = QH$.

Every real square matrix can be factored into $A = QH$, where Q is *orthogonal* and H is *symmetric positive semidefinite*. If A is invertible, H is positive definite.

For the proof we just insert $V^T V = I$ into the middle of the SVD:

$$\text{Polar decomposition} \quad A = U \Sigma V^T = (UV^T)(V \Sigma V^T) = (Q)(H). \quad (5)$$

The first factor UV^T is Q . The product of orthogonal matrices is orthogonal. The second factor $V \Sigma V^T$ is H . It is positive semidefinite because its eigenvalues are in Σ . If A is invertible then Σ and H are also invertible. **H is the symmetric positive definite square root of $A^T A$.** Equation (5) says that $H^2 = V \Sigma^2 V^T = A^T A$.

There is also a polar decomposition $A = KQ$ in the reverse order. Q is the same but now $K = U \Sigma U^T$. This is the symmetric positive definite square root of AA^T .

Example 3 Find the polar decomposition $A = QH$ from its SVD in Section 6.7:

$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{2} & \\ & 2\sqrt{2} \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = U \Sigma V^T.$$

Solution The orthogonal part is $Q = UV^T$. The positive definite part is $H = V \Sigma V^T$. This is also $H = Q^{-1}A$ which is $Q^T A$ because Q is orthogonal:

$$\begin{array}{ll} \text{Orthogonal} & Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \\ \text{Positive definite} & H = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 3/\sqrt{2} \end{bmatrix}. \end{array}$$

The polar decomposition just splits the key equation $Av_i = \sigma_i u_i$ into two steps. The “ H ” part multiplies v_i by σ_i . The “ Q ” part swings v_i around into u_i .

By choosing good bases, A multiplies v_i in the row space to give $\sigma_i u_i$ in the column space. A^{-1} must do the opposite! If $Av = \sigma u$ then $A^{-1}u = v/\sigma$. The singular values of A^{-1} are $1/\sigma$, just as the eigenvalues of A^{-1} are $1/\lambda$. The bases are reversed. The u 's are in the row space of A^{-1} , the v 's are in the column space.

Pseudoinverse

$$A^+ = V \Sigma^+ U^T = \begin{bmatrix} v_1 \cdots v_r \cdots v_n \\ n \text{ by } n \end{bmatrix} \begin{bmatrix} \sigma_1^{-1} & & \\ & \ddots & \\ & & \sigma_r^{-1} \\ & & & 0 \end{bmatrix} \begin{bmatrix} u_1 \cdots u_r \cdots u_m \\ m \text{ by } m \end{bmatrix}^T$$
$$A^+ u_i = \frac{1}{\sigma_i} v_i \quad \text{for } i \leq r \quad \text{and} \quad A^+ u_i = \mathbf{0} \quad \text{for } i > r.$$

Notice the pseudoinverse Σ^+ of the diagonal matrix Σ . Each σ is replaced by σ^{-1} . The product $\Sigma^+\Sigma$ is as near to the identity as we can get (it is a projection matrix, $\Sigma^+\Sigma$ is partly I and partly 0). We get r 1's. We can't do anything about the zero rows and columns. This example has $\sigma_1 = 2$ and $\sigma_2 = 3$:

$$\Sigma^+ \Sigma = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}.$$

The pseudoinverse A^+ is the n by m matrix that makes AA^+ and A^+A into projections:

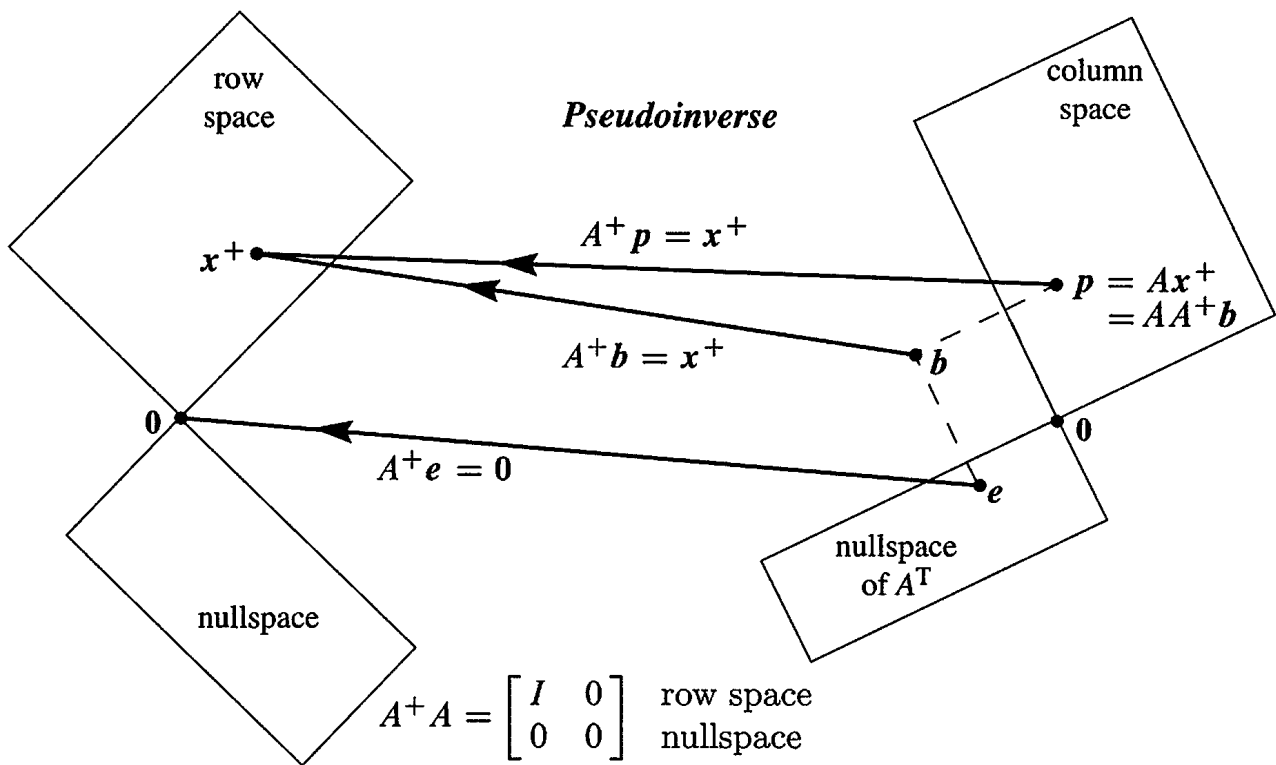


Figure 7.4: Ax^+ in the column space goes back to $A^+Ax^+ = x^+$ in the row space.

Trying for

$$AA^{-1} = A^{-1}A = I$$

$AA^+ =$ projection matrix onto the column space of A
 $A^+A =$ projection matrix onto the row space of A

Example 4 Find the pseudoinverse of $A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$. This matrix is not invertible. The rank is 1. The only singular value is $\sqrt{10}$. That is inverted to $1/\sqrt{10}$ in Σ^+ :

$$A^+ = V\Sigma^+U^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{10} & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}.$$

A^+ also has rank 1. Its column space is the row space of A . When A takes $(1, 1)$ in the row space to $(4, 2)$ in the column space, A^+ does the reverse. $A^+(4, 2) = (1, 1)$.

Every rank one matrix is a column times a row. With unit vectors u and v , that is $A = \sigma uv^T$. Then the best inverse of a rank one matrix is $A^+ = vu^T/\sigma$. The product AA^+ is uu^T , the projection onto the line through u . The product A^+A is vv^T .

Application to least squares Chapter 4 found the best solution $\hat{x}$ to an unsolvable system $Ax = b$. The key equation is $A^T A \hat{x} = A^T b$, with the assumption that $A^T A$ is invertible. The zero vector was alone in the nullspace.

Now A may have dependent columns ($\text{rank} < n$). There can be many solutions to $A^T A \hat{x} = A^T b$. One solution is $x^+ = A^+ b$ from the pseudoinverse. We can check that

$A^T A A^+ b$, is $A^T b$, because Figure 7.4 shows that $e = b - A A^+ b$ is the part of b in the nullspace of A^T . Any vector in the nullspace of A could be added to x^+ , to give another solution $\hat{x}$ to $A^T A \hat{x} = A^T b$. But x^+ will be shorter than any other $\hat{x}$ (Problem 16):

The shortest least squares solution to $Ax = b$ is $x^+ = A^+ b$.

The pseudoinverse A^+ and this best solution x^+ are essential in statistics, because experiments often have a matrix A with **dependent columns**.

■ REVIEW OF THE KEY IDEAS ■

1. Diagonalization $S^{-1}AS = \Lambda$ is the same as a change to the eigenvector basis.
2. The SVD chooses an input basis of v 's and an output basis of u 's. Those orthonormal bases diagonalize A . This is $Av_i = \sigma_i u_i$, and in matrix form $A = U\Sigma V^T$.
3. Polar decomposition factors A into QH , rotation UV^T times stretching $V\Sigma V^T$.
4. The pseudoinverse $A^+ = V\Sigma^+U^T$ transforms the column space of A back to its row space. A^+A is the identity on the row space (and zero on the nullspace).

■ WORKED EXAMPLES ■

7.3 A If A has rank n (full column rank) then it has a *left inverse* $C = (A^T A)^{-1} A^T$. This matrix C gives $CA = I$. Explain why the pseudoinverse is $A^+ = C$ in this case. If A has rank m (full row rank) then it has a *right inverse* B with $B = A^T (AA^T)^{-1}$. Then $AB = I$. Explain why $A^+ = B$ in this case.

Find B for A_1 and find C for A_2 . Find A^+ for all three matrices A_1, A_2, A_3 :

$$A_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad A_2 = \begin{bmatrix} 2 & 2 \end{bmatrix} \quad A_3 = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}.$$

Solution If A has rank n (independent columns) then $A^T A$ is invertible—this is a key point of Section 4.2. Certainly $C = (A^T A)^{-1} A^T$ multiplies A to give $CA = I$. In the opposite order, $AC = A(A^T A)^{-1} A^T$ is the projection matrix (Section 4.2 again) onto the column space. So C meets the requirements to be A^+ : CA and AC are projections.

If A has rank m (full row rank) then AA^T is invertible. Certainly A multiplies $B = A^T (AA^T)^{-1}$ to give $AB = I$. In the opposite order, $BA = A^T (AA^T)^{-1} A$ is the projection matrix onto the row space. So B is the pseudoinverse A^+ with rank m .

The example A_1 has full column rank (for C) and A_2 has full row rank (for B):

$$A_1^+ = (A_1^T A_1)^{-1} A_1^T = \frac{1}{\sqrt{8}} \begin{bmatrix} 2 & 2 \end{bmatrix} \quad A_2^+ = A_2^T (A_2 A_2^T)^{-1} = \frac{1}{\sqrt{8}} \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Notice $A_1^+ A_1 = [1]$ and $A_2 A_2^+ = [1]$. But A_3 (rank 1) has no left or right inverse. **Its rank is not full.** Its pseudoinverse is $A_3^+ = \sigma_1^{-1} v_1 u_1^T = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} / 4$.

Problem Set 7.3

Problems 1–4 compute and use the SVD of a particular matrix (not invertible).

- 1 (a) Compute $A^T A$ and its eigenvalues and unit eigenvectors v_1 and v_2 . Find σ_1 .

Rank one matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$

- (b) Compute AA^T and its eigenvalues and unit eigenvectors u_1 and u_2 .

- (c) Verify that $Av_1 = \sigma_1 u_1$. Put numbers into the SVD:

$$A = U \Sigma V^T \quad \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^T.$$

- 2 (a) From the u 's and v 's in Problem 1 write down orthonormal bases for the four fundamental subspaces of this matrix A .
- (b) Describe all matrices that have those same four subspaces. Multiples of A ?
- 3 From U , V , and Σ in Problem 1 find the orthogonal matrix $Q = UV^T$ and the symmetric matrix $H = V\Sigma V^T$. Verify the polar decomposition $A = QH$. This H is only semidefinite because _____. Test $H^2 = A$.
- 4 Compute the pseudoinverse $A^+ = V\Sigma^+U^T$. The diagonal matrix Σ^+ contains $1/\sigma_1$. Rename the four subspaces (for A) in Figure 7.4 as four subspaces for A^+ . Compute the projections $P_{\text{row}} = A^+A$ and $P_{\text{column}} = AA^+$.

Problems 5–9 are about the SVD of an invertible matrix.

- 5 Compute $A^T A$ and its eigenvalues and unit eigenvectors v_1 and v_2 . What are the singular values σ_1 and σ_2 for this matrix A ?

$$A = \begin{bmatrix} 3 & 3 \\ -1 & 1 \end{bmatrix}.$$

- 6 AA^T has the same eigenvalues σ_1^2 and σ_2^2 as $A^T A$. Find unit eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$. Put numbers into the SVD:

$$A = \begin{bmatrix} 3 & 3 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix}^T.$$

- 7 In Problem 6, multiply columns times rows to show that $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T$. Prove from $A = U \Sigma V^T$ that every matrix of rank r is the sum of r matrices of rank one.
- 8 From U , V , and Σ find the orthogonal matrix $Q = UV^T$ and the symmetric matrix $K = U \Sigma U^T$. Verify the polar decomposition in reverse order $A = KQ$.
- 9 The pseudoinverse of this A is the same as _____ because _____.

Problems 10–11 compute and use the SVD of a 1 by 3 rectangular matrix.

- 10 Compute $A^T A$ and AA^T and their eigenvalues and unit eigenvectors when the matrix is $A = \begin{bmatrix} 3 & 4 & 0 \end{bmatrix}$. What are the singular values of A ?
- 11 Put numbers into the singular value decomposition of A :

$$A = \begin{bmatrix} 3 & 4 & 0 \end{bmatrix} = [\mathbf{u}_1] \begin{bmatrix} \sigma_1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix}^T.$$

Put numbers into the pseudoinverse $V \Sigma^+ U^T$ of A . Compute AA^+ and $A^+ A$:

$$\text{Pseudoinverse } A^+ = \begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} \begin{bmatrix} 1/\sigma_1 \\ 0 \\ 0 \end{bmatrix} [\mathbf{u}_1]^T.$$

- 12 What is the only 2 by 3 matrix that has no pivots and no singular values? What is Σ for that matrix? A^+ is the zero matrix, but what shape?
- 13 If $\det A = 0$ why is $\det A^+ = 0$? If A has rank r , why does A^+ have rank r ?
- 14 When are the factors in $U \Sigma V^T$ the same as in $Q \Lambda Q^T$? The eigenvalues λ_i must be positive, to equal the σ_i . Then A must be _____ and positive _____.

Problems 15–18 bring out the main properties of A^+ and $\mathbf{x}^+ = A^+ \mathbf{b}$.

- 15 All matrices in this problem have rank one. The vector $\mathbf{b}$ is (b_1, b_2) .

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} .2 & .1 \\ .2 & .1 \end{bmatrix} \quad AA^T = \begin{bmatrix} .8 & .4 \\ .4 & .2 \end{bmatrix} \quad A^T A = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$$

- (a) The equation $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ has many solutions because $A^T A$ is _____.
- (b) Verify that $\mathbf{x}^+ = A^+ \mathbf{b} = (.2b_1 + .1b_2, .2b_1 + .1b_2)$ solves $A^T A \mathbf{x}^+ = A^T \mathbf{b}$.

(c) Add $(1, -1)$ to that $\mathbf{x}^+$ to get another solution to $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. Show that $\|\hat{\mathbf{x}}\|^2 = \|\mathbf{x}^+\|^2 + 2$, and $\mathbf{x}^+$ is shorter.

16 *The vector $\mathbf{x}^+ = A^+ \mathbf{b}$ is the shortest possible solution to $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.* Reason: The difference $\hat{\mathbf{x}} - \mathbf{x}^+$ is in the nullspace of $A^T A$. This is also the nullspace of A , orthogonal to $\mathbf{x}^+$. Explain how it follows that $\|\hat{\mathbf{x}}\|^2 = \|\mathbf{x}^+\|^2 + \|\hat{\mathbf{x}} - \mathbf{x}^+\|^2$.

17 Every $\mathbf{b}$ in $\mathbf{R}^m$ is $\mathbf{p} + \mathbf{e}$. This is the column space part plus the left nullspace part. Every $\mathbf{x}$ in $\mathbf{R}^n$ is $\mathbf{x}_r + \mathbf{x}_n = (\text{row space part}) + (\text{nullspace part})$. Then

$$AA^+ \mathbf{p} = \underline{\hspace{2cm}} \quad AA^+ \mathbf{e} = \underline{\hspace{2cm}} \quad A^+ A \mathbf{x}_r = \underline{\hspace{2cm}} \quad A^+ A \mathbf{x}_n = \underline{\hspace{2cm}}$$

18 Find A^+ and $A^+ A$ and AA^+ and $\mathbf{x}^+$ for this 2 by 1 matrix and these $\mathbf{b}$:

$$A = \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} .6 & -.8 \\ .8 & .6 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \end{bmatrix} [1] \quad \mathbf{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}.$$

19 A general 2 by 2 matrix A is determined by four numbers. If triangular, it is determined by three. If diagonal, by two. If a rotation, by one. An eigenvector, by one. Check that the total count is four for each factorization of A :

$$\text{Four numbers in } LU \quad LDU \quad QR \quad U\Sigma V^T \quad SAS^{-1}.$$

20 Following Problem 19, check that LDL^T and $Q\Lambda Q^T$ are determined by *three* numbers. This is correct because the matrix A is now $\underline{\hspace{2cm}}$.

21 From $A = U\Sigma V^T$ and $A^+ = V\Sigma^+ U^T$ explain these splittings into rank 1:

$$A = \sum_1^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T \quad A^+ = \sum_1^r \frac{\mathbf{v}_i \mathbf{u}_i^T}{\sigma_i} \quad A^+ A = \sum_1^r \mathbf{v}_i \mathbf{v}_i^T \quad AA^+ = \sum_1^r \mathbf{u}_i \mathbf{u}_i^T$$

Challenge Problems

22 This problem looks for all matrices A with a given column space in $\mathbf{R}^m$ and a given row space in $\mathbf{R}^n$. Suppose $\mathbf{c}_1, \dots, \mathbf{c}_r$ and $\mathbf{b}_1, \dots, \mathbf{b}_r$ are bases for those two spaces. Make them columns of C and B . The goal is to show that $A = \mathbf{CMB}^T$ for an r by r invertible matrix M . Hint: Start from $A = U\Sigma V^T$. A must have this form:

The first r columns of U and V must be connected to C and B by invertible matrices, because they contain bases for the same column space and row space.

23 A pair of singular vectors $\mathbf{v}$ and $\mathbf{u}$ will satisfy $A\mathbf{v} = \sigma\mathbf{u}$ and $A^T\mathbf{u} = \sigma\mathbf{v}$. This means that the double vector $\mathbf{x} = \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix}$ is an eigenvector of what symmetric block matrix? What is the eigenvalue? The SVD of A is equivalent to the diagonalization of that symmetric block matrix.

Chapter 8

Applications

8.1 Matrices in Engineering

This section will show how engineering problems produce symmetric matrices K (often K is positive definite). The “linear algebra reason” for symmetry and positive definiteness is their form $K = A^T A$ and $K = A^T C A$. The “physical reason” is that the expression $\frac{1}{2} \mathbf{u}^T K \mathbf{u}$ represents *energy*—and energy is never negative. The matrix C , often diagonal, contains positive physical constants like conductance or stiffness or diffusivity.

Our first examples come from mechanical and civil and aeronautical engineering. K is the *stiffness matrix*, and $K^{-1} \mathbf{f}$ is the structure’s response to forces $\mathbf{f}$ from outside. Section 8.2 turns to electrical engineering—the matrices come from networks and circuits. The exercises involve chemical engineering and I could go on! Economics and management and engineering design come later in this chapter (there the key is optimization).

Engineering leads to linear algebra in two ways, directly and indirectly:

Direct way The physical problem has only a finite number of pieces. The laws connecting their position or velocity are *linear* (movement is not too big or too fast). The laws are expressed by *matrix equations*.

Indirect way The physical system is “continuous”. Instead of individual masses, the mass density and the forces and the velocities are functions of x or x, y or x, y, z . The laws are expressed by *differential equations*. *To find accurate solutions we approximate by finite difference equations or finite element equations.*

Both ways produce matrix equations and linear algebra. I really believe that you cannot do modern engineering without matrices.

Here we present equilibrium equations $K \mathbf{u} = \mathbf{f}$. With motion, $M d^2 \mathbf{u} / dt^2 + K \mathbf{u} = \mathbf{f}$ becomes dynamic. Then we use eigenvalues from $K \mathbf{x} = \lambda M \mathbf{x}$, or finite differences.

Before explaining the physical examples, may I write down the matrices? The tridiagonal K_0 appears many times in this textbook. Now we will see its applications. These matrices are all symmetric, and the first four are positive definite:

$$K_0 = A_0^T A_0 = \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & \end{bmatrix} \quad A_0^T C_0 A_0 = \begin{bmatrix} c_1 + c_2 & -c_2 & & \\ -c_2 & c_2 + c_3 & -c_3 & \\ & -c_3 & c_3 + c_4 & \end{bmatrix}$$

Fixed-fixed

Spring constants included

$$K_1 = A_1^T A_1 = \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 1 & \end{bmatrix} \quad A_1^T C_1 A_1 = \begin{bmatrix} c_1 + c_2 & -c_2 & & \\ -c_2 & c_2 + c_3 & -c_3 & \\ & -c_3 & c_3 & \end{bmatrix}$$

Fixed-free

Spring constants included

$$K_{\text{singular}} = \begin{bmatrix} 1 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 1 & \end{bmatrix} \quad K_{\text{circular}} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Free-free

The matrices K_0 , K_1 , K_{singular} , and K_{circular} have $C = I$ for simplicity. This means that all the “spring constants” are $c_i = 1$. We included $A_0^T C_0 A_0$ and $A_1^T C_1 A_1$ to show how the spring constants enter the matrix (without changing its positive definiteness). Our first goal is to show where these stiffness matrices come from.

A Line of Springs

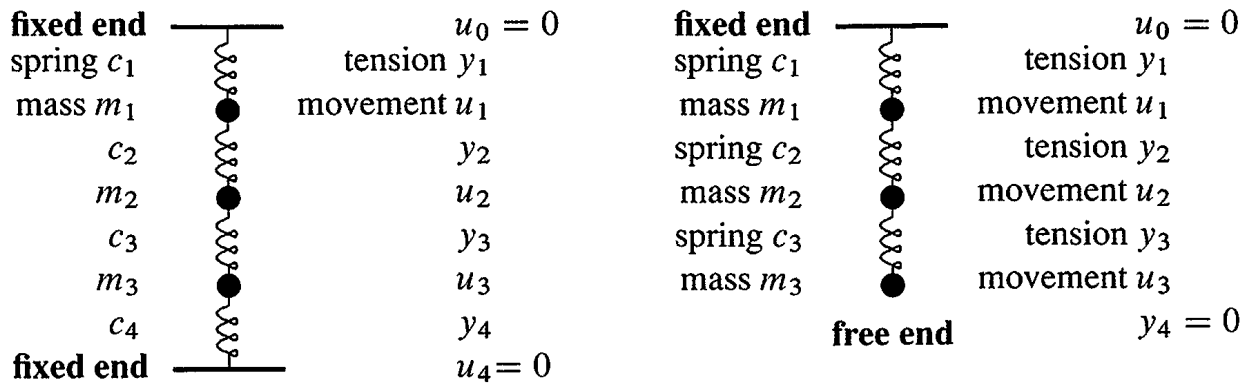
Figure 8.1 shows three masses m_1, m_2, m_3 connected by a line of springs. One case has four springs, with top and bottom fixed. The fixed-free case has only three springs; the lowest mass hangs freely. The **fixed-fixed** problem will lead to K_0 and $A_0^T C_0 A_0$. The **fixed-free** problem will lead to K_1 and $A_1^T C_1 A_1$. A **free-free** problem, with no support at either end, produces the matrix K_{singular} .

We want equations for the mass movements u and the tensions (or compressions) y :

$$\begin{aligned} u &= (u_1, u_2, u_3) = \text{movements of the masses (down or up)} \\ y &= (y_1, y_2, y_3, y_4) \text{ or } (y_1, y_2, y_3) = \text{tensions in the springs} \end{aligned}$$

When a mass moves downward, its displacement is positive ($u_i > 0$). For the springs, tension is positive and compression is negative ($y_i < 0$). In tension, the spring is stretched so it pulls the masses inward. Each spring is controlled by its own Hooke’s Law $y = c e$: (*stretching force*) = (*spring constant*) times (*stretching distance*).

Our job is to link these one-spring equations $y = ce$ into a vector equation $Ku = f$ for the whole system. The force vector f comes from gravity. The gravitational constant g will multiply each mass to produce forces $f = (m_1 g, m_2 g, m_3 g)$.

Figure 8.1: Lines of springs and masses: **fixed-fixed** and **fixed-free** ends.

The real problem is to find the stiffness matrix (**fixed-fixed** and **fixed-free**). The best way to create K is in three steps, not one. Instead of connecting the movements u_i directly to the forces, it is much better to connect each vector to the next in this list:

$$\begin{aligned}
 u &= \text{Movements of } n \text{ masses} &= (u_1, \dots, u_n) \\
 e &= \text{Elongations of } m \text{ springs} &= (e_1, \dots, e_m) \\
 y &= \text{Internal forces in } m \text{ springs} &= (y_1, \dots, y_m) \\
 f &= \text{External forces on } n \text{ masses} &= (f_1, \dots, f_n)
 \end{aligned}$$

The framework that connects u to e to y to f looks like this:

$$\begin{array}{ccc}
 \boxed{u} & \boxed{f} & e = Au \quad A \text{ is } m \text{ by } n \\
 A \downarrow & \uparrow A^T & y = Ce \quad C \text{ is } m \text{ by } m \\
 \boxed{e} & \xrightarrow{C} \boxed{y} & f = A^T y \quad A^T \text{ is } n \text{ by } m
 \end{array}$$

We will write down the matrices A and C and A^T for the two examples, first with fixed ends and then with the lower end free. Forgive the simplicity of these matrices, it is their form that is so important. Especially the appearance of A together with A^T .

The *elongation* e is the stretching distance—how far the springs are extended. Originally there is no stretching—the system is lying on a table. When it becomes vertical and upright, gravity acts. The masses move down by distances u_1, u_2, u_3 . Each spring is stretched or compressed by $e_i = u_i - u_{i-1}$, the *difference in displacements of its ends*:

$$\begin{array}{lll}
 \text{Stretching of} & \text{First spring:} & e_1 = u_1 \quad (\text{the top is fixed so } u_0 = 0) \\
 \text{each spring} & \text{Second spring:} & e_2 = u_2 - u_1 \\
 & \text{Third spring:} & e_3 = u_3 - u_2 \\
 & \text{Fourth spring:} & e_4 = -u_3 \quad (\text{the bottom is fixed so } u_4 = 0)
 \end{array}$$

If both ends move the same distance, that spring is not stretched: $u_i = u_{i-1}$ and $e_i = 0$. The matrix in those four equations is a 4 by 3 *difference matrix* A , and $e = Au$:

$$\begin{array}{l} \text{Stretching} \\ \text{distances} \\ \text{(elongations)} \end{array} \quad e = Au \quad \text{is} \quad \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}. \quad (1)$$

The next equation $y = Ce$ connects spring elongation e with spring tension y . *This is Hooke's Law* $y_i = c_i e_i$ for each separate spring. It is the "constitutive law" that depends on the material in the spring. A soft spring has small c , so a moderate force y can produce a large stretching e . Hooke's linear law is nearly exact for real springs, before they are overstretched and the material becomes plastic.

Since each spring has its own law, the matrix in $y = Ce$ is a diagonal matrix C :

$$\begin{array}{l} \text{Hooke's} \\ \text{Law} \\ y = Ce \end{array} \quad \begin{array}{l} y_1 = c_1 e_1 \\ y_2 = c_2 e_2 \\ y_3 = c_3 e_3 \\ y_4 = c_4 e_4 \end{array} \quad \text{is} \quad \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} c_1 & & & \\ & c_2 & & \\ & & c_3 & \\ & & & c_4 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix} \quad (2)$$

Combining $e = Au$ with $y = Ce$, the spring forces are $y = CAu$.

Finally comes the *balance equation*, the most fundamental law of applied mathematics. The internal forces from the springs balance the external forces on the masses. Each mass is pulled or pushed by the spring force y_j above it. From below it feels the spring force y_{j+1} plus f_j from gravity. Thus $y_j = y_{j+1} + f_j$ or $f_j = y_j - y_{j+1}$:

$$\begin{array}{l} \text{Force} \\ \text{balance} \\ f = A^T y \end{array} \quad \begin{array}{l} f_1 = y_1 - y_2 \\ f_2 = y_2 - y_3 \\ f_3 = y_3 - y_4 \end{array} \quad \text{is} \quad \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (3)$$

That matrix is A^T . The equation for balance of forces is $f = A^T y$. Nature transposes the rows and columns of the $e - u$ matrix to produce the $f - y$ matrix. This is the beauty of the framework, that A^T appears along with A . The three equations combine into $Ku = f$, where the *stiffness matrix* is $K = A^T CA$:

$$\left\{ \begin{array}{l} e = Au \\ y = Ce \\ f = A^T y \end{array} \right\} \quad \text{combine into} \quad A^T CAu = f \quad \text{or} \quad Ku = f.$$

In the language of elasticity, $e = Au$ is the **kinematic** equation (for displacement). The force balance $f = A^T y$ is the **static** equation (for equilibrium). The **constitutive law** is $y = Ce$ (from the material). Then $A^T CA$ is n by $n = (n \text{ by } m)(m \text{ by } m)(m \text{ by } n)$.

Finite element programs spend major effort on assembling $K = A^T CA$ from thousands of smaller pieces. We find K for four springs (**fixed-fixed**) by multiplying A^T times CA :

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} c_1 & 0 & 0 \\ -c_2 & c_2 & 0 \\ 0 & -c_3 & c_3 \\ 0 & 0 & -c_4 \end{bmatrix} = \begin{bmatrix} c_1 + c_2 & -c_2 & 0 \\ -c_2 & c_2 + c_3 & -c_3 \\ 0 & -c_3 & c_3 + c_4 \end{bmatrix}$$

If all springs are identical, with $c_1 = c_2 = c_3 = c_4 = 1$, then $C = I$. The stiffness matrix reduces to $A^T A$. It becomes the special $-1, 2, -1$ matrix:

$$\text{With } C = I \quad K_0 = A_0^T A_0 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}. \quad (4)$$

Note the difference between $A^T A$ from engineering and LL^T from linear algebra. The matrix A from four springs is 4 by 3. The triangular matrix L from elimination is square. The stiffness matrix K is assembled from $A^T A$, and then broken up into LL^T . One step is applied mathematics, the other is computational mathematics. Each K is built from rectangular matrices and factored into square matrices.

May I list some properties of $K = A^T C A$? You know almost all of them:

1. K is **tridiagonal**, because mass 3 is not connected to mass 1.
2. K is **symmetric**, because C is symmetric and A^T comes with A .
3. K is **positive definite**, because $c_i > 0$ and A has **independent columns**.
4. K^{-1} is a full matrix in equation (5) with **all positive entries**.

That last property leads to an important fact about $u = K^{-1} f$: *If all forces act downwards ($f_j > 0$) then all movements are downwards ($u_j > 0$).* Notice that “positiveness” is different from “positive definiteness”. Here K^{-1} is positive (K is not). Both K and K^{-1} are positive definite.

Example 1 Suppose all $c_i = c$ and $m_j = m$. Find the movements u and tensions y .

All springs are the same and all masses are the same. But all movements and elongations and tensions will *not* be the same. K^{-1} includes $\frac{1}{c}$ because $A^T C A$ includes c :

$$u = K^{-1} f = \frac{1}{4c} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} mg \\ mg \\ mg \end{bmatrix} = \frac{mg}{c} \begin{bmatrix} 3/2 \\ 2 \\ 3/2 \end{bmatrix} \quad (5)$$

The displacement u_2 , for the mass in the middle, is greater than u_1 and u_3 . The units are correct: the force mg divided by force per unit length c gives a length u . Then

$$e = Au = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \frac{mg}{c} \begin{bmatrix} \frac{3}{2} \\ 2 \\ \frac{3}{2} \end{bmatrix} = \frac{mg}{c} \begin{bmatrix} 3/2 \\ 1/2 \\ -1/2 \\ -3/2 \end{bmatrix}.$$

Those elongations add to zero because the ends of the line are fixed. (The sum $u_1 + (u_2 - u_1) + (u_3 - u_2) + (-u_3)$ is certainly zero.) For each spring force y_i we just multiply e_i by c . So y_1, y_2, y_3, y_4 are $\frac{3}{2}mg, \frac{1}{2}mg, -\frac{1}{2}mg, -\frac{3}{2}mg$. The upper two springs are stretched, the lower two springs are compressed.

Notice how u, e, y are computed in that order. We assembled $K = A^T C A$ from rectangular matrices. To find $u = K^{-1} f$, we work with the whole matrix and not its three pieces! The rectangular matrices A and A^T do not have (two-sided) inverses.

Warning: Normally you cannot write $K^{-1} = A^{-1}C^{-1}(A^T)^{-1}$.

The three matrices are mixed together by A^TCA , and they cannot easily be untangled. In general, $A^Ty = f$ has many solutions. And four equations $Au = e$ would usually have no solution with three unknowns. But A^TCA gives the correct solution to all three equations in the framework. Only when $m = n$ and the matrices are square can we go from $y = (A^T)^{-1}f$ to $e = C^{-1}y$ to $u = A^{-1}e$. We will see that now.

Fixed End and Free End

Remove the fourth spring. All matrices become 3 by 3. The pattern does not change! The matrix A loses its fourth row and (of course) A^T loses its fourth column. The new stiffness matrix K_1 becomes a product of square matrices:

$$A_1^T C_1 A_1 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 & & \\ & c_2 & \\ & & c_3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

The missing column of A^T and row of A multiplied the missing c_4 . So the quickest way to find the new A^TCA is to set $c_4 = 0$ in the old one:

$$\begin{array}{l} \text{FIXED} \\ \text{FREE} \end{array} \quad K_1 = A_1^T C_1 A_1 = \begin{bmatrix} c_1 + c_2 & -c_2 & 0 \\ -c_2 & c_2 + c_3 & -c_3 \\ 0 & -c_3 & c_3 \end{bmatrix}. \quad (6)$$

If $c_1 = c_2 = c_3 = 1$ and $C = I$, this is the $-1, 2, -1$ tridiagonal matrix, except the last entry is 1 instead of 2. The spring at the bottom is free.

Example 2 All $c_i = c$ and all $m_j = m$ in the fixed-free hanging line of springs. Then

$$K_1 = c \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \quad \text{and} \quad K_1^{-1} = \frac{1}{c} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

The forces mg from gravity are the same. But the movements change from the previous example because the stiffness matrix has changed:

$$u = K_1^{-1} f = \frac{1}{c} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} mg \\ mg \\ mg \end{bmatrix} = \frac{mg}{c} \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix}.$$

Those movements are greater in this fixed-free case. The number 3 appears in u_1 because all three masses are pulling the first spring down. The next mass moves by that 3 plus an additional 2 from the masses below it. The third mass drops even more ($3 + 2 + 1 = 6$). The elongations $e = Au$ in the springs display those numbers 3, 2, 1:

$$e = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \frac{mg}{c} \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix} = \frac{mg}{c} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}.$$

Multiplying by c , the forces y in the three springs are $3mg$ and $2mg$ and mg . And the special point of square matrices is that y can be found directly from f ! The balance equation $A^T y = f$ determines y immediately, because $m = n$ and A^T is square. We are allowed to write $(A^T C A)^{-1} = A^{-1} C^{-1} (A^T)^{-1}$:

$$y = (A^T)^{-1} f \text{ is } \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} mg \\ mg \\ mg \end{bmatrix} = \begin{bmatrix} 3mg \\ 2mg \\ 1mg \end{bmatrix}.$$

Two Free Ends: K is Singular

The first line of springs in Figure 8.2 is free at *both ends*. This means trouble (the whole line can move). The matrix A is 2 by 3, short and wide. Here is $e = Au$:

$$\text{FREE-FREE} \quad \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} u_2 - u_1 \\ u_3 - u_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}. \quad (7)$$

Now there is a nonzero solution to $Au = 0$. *The masses can move with no stretching of the springs*. The whole line can shift by $u = (1, 1, 1)$ and this leaves $e = (0, 0)$. A has *dependent columns* and the vector $(1, 1, 1)$ is in its nullspace:

$$Au = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \text{no stretching}. \quad (8)$$

$Au = 0$ certainly leads to $A^T C Au = 0$. So $A^T C A$ is only *positive semidefinite*, without c_1 and c_4 . The pivots will be c_2 and c_3 and *no third pivot*. The rank is only 2:

$$\begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} c_2 & \\ & c_3 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} c_2 & -c_2 & 0 \\ -c_2 & c_2 + c_3 & -c_3 \\ 0 & -c_3 & c_3 \end{bmatrix} \quad (9)$$

Two eigenvalues will be positive but $x = (1, 1, 1)$ is an eigenvector for $\lambda = 0$. We can solve $A^T C Au = f$ only for special vectors f . The forces have to add to $f_1 + f_2 + f_3 = 0$, or the whole line of springs (with both ends free) will take off like a rocket.

Circle of Springs

A third spring will complete the circle from mass 3 back to mass 1. This doesn't make K invertible—the new matrix is still singular. That stiffness matrix K_{circular} is not tridiagonal, but it is symmetric (always) and *semidefinite*:

$$A_{\text{circular}}^T A_{\text{circular}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}. \quad (10)$$

The only pivots are 2 and $\frac{3}{2}$. The eigenvalues are 3 and 3 and 0. The determinant is zero. The nullspace still contains $x = (1, 1, 1)$, when all the masses move together.

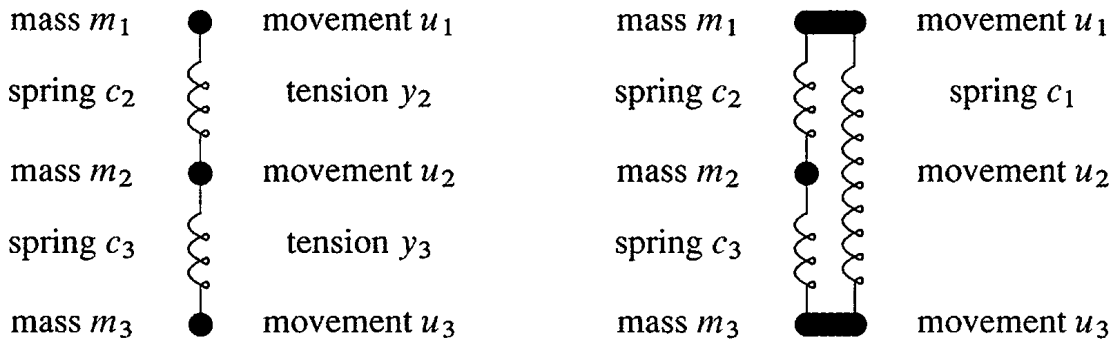


Figure 8.2: **Free-free ends:** A line of springs and a “circle” of springs: *Singular* K ’s. The masses can move without stretching the springs so $Au = 0$ has nonzero solutions.

This movement vector $(1, 1, 1)$ is in the nullspace of A_{circular} and K_{circular} , even after the diagonal matrix C of spring constants is included: the springs are not stretched.

$$(A^T C A)_{\text{circular}} = \begin{bmatrix} c_1 + c_2 & -c_2 & -c_1 \\ -c_2 & c_2 + c_3 & -c_3 \\ -c_1 & -c_3 & c_3 + c_1 \end{bmatrix}. \quad (11)$$

Continuous Instead of Discrete

Matrix equations are discrete. Differential equations are continuous. We will see the differential equation that corresponds to the tridiagonal $-1, 2, -1$ matrix $A^T A$. And it is a pleasure to see the boundary conditions that go with K_0 and K_1 .

The matrices A and A^T correspond to the derivatives d/dx and $-d/dx$! Remember that $e = Au$ took differences $u_i - u_{i-1}$, and $f = A^T y$ took differences $y_i - y_{i+1}$. Now the springs are infinitesimally short, and those differences become derivatives:

$$\frac{u_i - u_{i-1}}{\Delta x} \text{ is like } \frac{du}{dx} \quad \frac{y_i - y_{i+1}}{\Delta x} \text{ is like } -\frac{dy}{dx}$$

The factor Δx didn’t appear earlier—we imagined the distance between masses was 1. To approximate a continuous solid bar, we take many more masses (smaller and closer). Let me jump to the three steps A, C, A^T in the continuous model, when there is stretching and Hooke’s Law and force balance at every point x :

$$e(x) = Au = \frac{du}{dx} \quad y(x) = c(x)e(x) \quad A^T y = -\frac{dy}{dx} = f(x)$$

Combining those equations into $A^T C A u(x) = f(x)$, we have a differential equation not a matrix equation. The line of springs becomes an elastic bar:

$$\text{Solid Elastic Bar} \quad A^T C A u(x) = f(x) \quad \text{is} \quad -\frac{d}{dx} \left(c(x) \frac{du}{dx} \right) = f(x) \quad (12)$$

$A^T A$ corresponds to a second derivative. A is a “difference matrix” and $A^T A$ is a “second difference matrix”. **The matrix has $-1, 2, -1$ and the equation has $-d^2u/dx^2$:**

$$-u_{i+1} + 2u_i - u_{i-1} \text{ is a second difference} \quad -\frac{d^2u}{dx^2} \text{ is a second derivative.}$$

Now we see why this symmetric matrix is a favorite. When we meet a first derivative du/dx , we have three choices (*forward, backward, and centered differences*):

$$\frac{du}{dx} \simeq \frac{u(x + \Delta x) - u(x)}{\Delta x} \quad \text{or} \quad \frac{u(x) - u(x - \Delta x)}{\Delta x} \quad \text{or} \quad \frac{u(x + \Delta x) - u(x - \Delta x)}{2\Delta x}.$$

When we meet d^2u/dx^2 , the natural choice is $u(x + \Delta x) - 2u(x) + u(x - \Delta x)$, divided by $(\Delta x)^2$. *Why reverse these signs to $-1, 2, -1$?* Because the positive definite matrix has $+2$ on the diagonal. First derivatives are *antisymmetric*; the transpose has a minus sign. So second differences are negative definite, and we change to $-d^2u/dx^2$.

We have moved from vectors to functions. Scientific computing moves the other way. It starts with a differential equation like (12). Sometimes there is a formula for the solution $u(x)$, more often not. In reality we *create* the discrete matrix K by approximating the continuous problem. Watch how the boundary conditions on u come in! By missing u_0 we treat it (correctly) as zero:

$$\begin{array}{l} \text{FIXED} \\ \text{FIXED} \end{array} \quad Au = \frac{1}{\Delta x} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \approx \frac{du}{dx} \quad \text{with} \quad \begin{array}{l} u_0 = 0 \\ u_4 = 0 \end{array} \quad (13)$$

Fixing the top end gives the boundary condition $u_0 = 0$. What about the free end, when the bar hangs in the air? Row 4 of A is gone and so is u_4 . The boundary condition must come from A^T . It is the missing y_4 that we are treating (correctly) as zero:

$$\begin{array}{l} \text{FIXED} \\ \text{FREE} \end{array} \quad A^T y = \frac{1}{\Delta x} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \approx -\frac{dy}{dx} \quad \text{with} \quad \begin{array}{l} u_0 = 0 \\ y_4 = 0 \end{array} \quad (14)$$

The boundary condition $y_4 = 0$ at the free end becomes $du/dx = 0$, since $y = Au$ corresponds to du/dx . The force balance $A^T y = f$ at that end (in the air) is $0 = 0$. The last row of $K_1 u = f$ has entries $-1, 1$ to reflect this condition $du/dx = 0$.

May I summarize this section? I hope this example will help you turn calculus into linear algebra, replacing differential equations by difference equations. If your step Δx is small enough, you will have a totally satisfactory solution.

$$\text{The equation is} \quad -\frac{d}{dx} \left(c(x) \frac{du}{dx} \right) = f(x) \quad \text{with} \quad u(0) = 0 \quad \text{and} \quad \left[u(1) \text{ or } \frac{du}{dx}(1) \right] = 0$$

Divide the bar into N pieces of length Δx . Replace du/dx by Au and $-dy/dx$ by $A^T y$. Now A and A^T include $1/\Delta x$. The end conditions are $u_0 = 0$ and $[u_N = 0 \text{ or } y_N = 0]$.

The three steps $-d/dx$ and $c(x)$ and d/dx correspond to A^T and C and A :

$$f = A^T y \text{ and } y = Ce \text{ and } e = Au \text{ give } A^T CAu = f.$$

This is a fundamental example in computational science and engineering. Our book concentrates on Step 3 in that process (linear algebra). Now we have taken Step 2.

1. Model the problem by a differential equation
2. Discretize the differential equation to a difference equation
3. Understand and solve the difference equation (and boundary conditions!)
4. Interpret the solution; visualize it; redesign if needed.

Numerical simulation has become a third branch of science, together with experiment and deduction. Designing the Boeing 777 was much less expensive on a computer than in a wind tunnel. Our discussion still has to move from ordinary to partial differential equations, and from linear to nonlinear.

The texts *Introduction to Applied Mathematics* and *Computational Science and Engineering* (Wellesley-Cambridge Press) develop this whole subject further—see the course page math.mit.edu/18085 with video lectures (also on ocw.mit.edu). The principles remain the same, and I hope this book helps you to see the framework behind the computations.

Problem Set 8.1

- 1 Show that $\det A_0^T C_0 A_0 = c_1 c_2 c_3 + c_1 c_3 c_4 + c_1 c_2 c_4 + c_2 c_3 c_4$. Find also $\det A_1^T C_1 A_1$ in the fixed-free example.
- 2 Invert $A_1^T C_1 A_1$ in the fixed-free example by multiplying $A_1^{-1} C_1^{-1} (A_1^T)^{-1}$.
- 3 In the free-free case when $A^T CA$ in equation (9) is singular, add the three equations $A^T CAu = f$ to show that we need $f_1 + f_2 + f_3 = 0$. Find a solution to $A^T CAu = f$ when the forces $f = (-1, 0, 1)$ balance themselves. Find all solutions!
- 4 Both end conditions for the free-free differential equation are $du/dx = 0$:

$$-\frac{d}{dx} \left(c(x) \frac{du}{dx} \right) = f(x) \quad \text{with} \quad \frac{du}{dx} = 0 \quad \text{at both ends.}$$

Integrate both sides to show that the force $f(x)$ must balance itself, $\int f(x) dx = 0$, or there is no solution. The complete solution is one particular solution $u(x)$ plus any constant. The constant corresponds to $u = (1, 1, 1)$ in the nullspace of $A^T CA$.

- 5 In the fixed-free problem, the matrix A is square and invertible. We can solve $A^T y = f$ separately from $Au = e$. Do the same for the differential equation:

$$\text{Solve } -\frac{dy}{dx} = f(x) \text{ with } y(1) = 0. \text{ Graph } y(x) \text{ if } f(x) = 1.$$

- 6 The 3 by 3 matrix $K_1 = A_1^T C_1 A_1$ in equation (6) splits into three “element matrices” $c_1 E_1 + c_2 E_2 + c_3 E_3$. Write down those pieces, one for each c . Show how they come from *column times row* multiplication of $A_1^T C_1 A_1$. This is how finite element stiffness matrices are actually assembled.
- 7 For five springs and four masses with both ends fixed, what are the matrices A and C and K ? With $C = I$ solve $Ku = \text{ones}(4)$.
- 8 Compare the solution $u = (u_1, u_2, u_3, u_4)$ in Problem 7 to the solution of the continuous problem $-u'' = 1$ with $u(0) = 0$ and $u(1) = 0$. The parabola $u(x)$ should correspond at $x = \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}$ to u —is there a $(\Delta x)^2$ factor to account for?
- 9 Solve the fixed-free problem $-u'' = mg$ with $u(0) = 0$ and $u'(1) = 0$. Compare $u(x)$ at $x = \frac{1}{3}, \frac{2}{3}, \frac{3}{3}$ with the vector $u = (3mg, 5mg, 6mg)$ in Example 2.
- 10 Suppose $c_1 = c_2 = c_3 = c_4 = 1$, $m_1 = 2$ and $m_2 = m_3 = 1$. Solve $A^T C A u = (2, 1, 1)$ for this fixed-fixed line of springs. Which mass moves the most (largest u)?
- 11 (MATLAB) Find the displacements $u(1), \dots, u(100)$ of 100 masses connected by springs all with $c = 1$. Each force is $f(i) = .01$. Print graphs of u with **fixed-fixed** and **fixed-free** ends. Note that $\text{diag}(\text{ones}(n, 1), d)$ is a matrix with n ones along diagonal d . This print command will graph a vector u :

```
plot(u, '+' );    xlabel('mass number');    ylabel('movement');    print
```

- 12 (MATLAB) Chemical engineering has a first derivative du/dx from fluid velocity as well as d^2u/dx^2 from diffusion. Replace du/dx by a *forward* difference, then a *centered* difference, then a *backward* difference, with $\Delta x = \frac{1}{8}$. Graph your three numerical solutions of

$$-\frac{d^2u}{dx^2} + 10 \frac{du}{dx} = 1 \quad \text{with } u(0) = u(1) = 0.$$

This *convection-diffusion equation* appears everywhere. It transforms to the Black-Scholes equation for option prices in mathematical finance.

Problem 12 is developed into the first MATLAB homework in my 18.085 course on Computational Science and Engineering at MIT. Videos on ocw.mit.edu.

8.2 Graphs and Networks

Over the years I have seen one model so often, and I found it so basic and useful, that I always put it first. The model consists of **nodes connected by edges**. This is called a **graph**.

Graphs of the usual kind display functions $f(x)$. Graphs of this node-edge kind lead to matrices. This section is about the **incidence matrix** of a graph—which tells how the n nodes are connected by the m edges. Normally $m > n$, there are more edges than nodes.

For any m by n matrix there are two fundamental subspaces in $\mathbf{R}^n$ and two in $\mathbf{R}^m$. They are the row spaces and nullspaces of A and A^T . Their *dimensions* are related by the most important theorem in linear algebra. The second part of that theorem is the *orthogonality* of the subspaces. Our goal is to show how examples from graphs illuminate the Fundamental Theorem of Linear Algebra.

We review the four subspaces (for any matrix). Then we construct a **directed graph** and its **incidence matrix**. The dimensions will be easy to discover. But we want the subspaces themselves—this is where orthogonality helps. It is essential to connect the subspaces to the graph they come from. By specializing to incidence matrices, the laws of linear algebra become Kirchhoff's laws. Please don't be put off by the words "current" and "voltage" and "Kirchhoff." These rectangular matrices are the best.

Every entry of an incidence matrix is 0 or 1 or -1 . This continues to hold during elimination. All pivots and multipliers are ± 1 . Therefore both factors in $A = LU$ also contain 0, 1, -1 . So do the nullspace matrices! All four subspaces have basis vectors with these exceptionally simple components. The matrices are not concocted for a textbook, they come from a model that is absolutely essential in pure and applied mathematics.

Here is a first incidence matrix. Notice -1 and 1 in each row. This matrix takes **differences in voltage**, across six edges of a graph. The voltages are x_1, x_2, x_3, x_4 at the four nodes in Figure 8.4—where we will construct this matrix A . Its echelon form is U :

$$\begin{array}{l} \text{Incidence} \\ \text{matrix} \\ \text{6 edges} \\ \text{4 nodes} \end{array} \quad A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad \text{reduces to} \quad U = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The nullspace of A and U is the line through $\mathbf{x} = (1, 1, 1, 1)$. The column spaces of A and U have dimension $r = 3$. The pivot rows are a basis for the row space.

Figure 8.3 shows more—the subspaces are orthogonal. **Every vector in the nullspace is perpendicular to every vector in the row space.** This comes directly from the m equations $A\mathbf{x} = \mathbf{0}$. For A and U above, $\mathbf{x} = (1, 1, 1, 1)$ is perpendicular to all rows and thus to the whole row space. Equal voltages produce no current!

I would like to review the Four Fundamental Subspaces before using them. The whole point will be to see their meaning on the network.

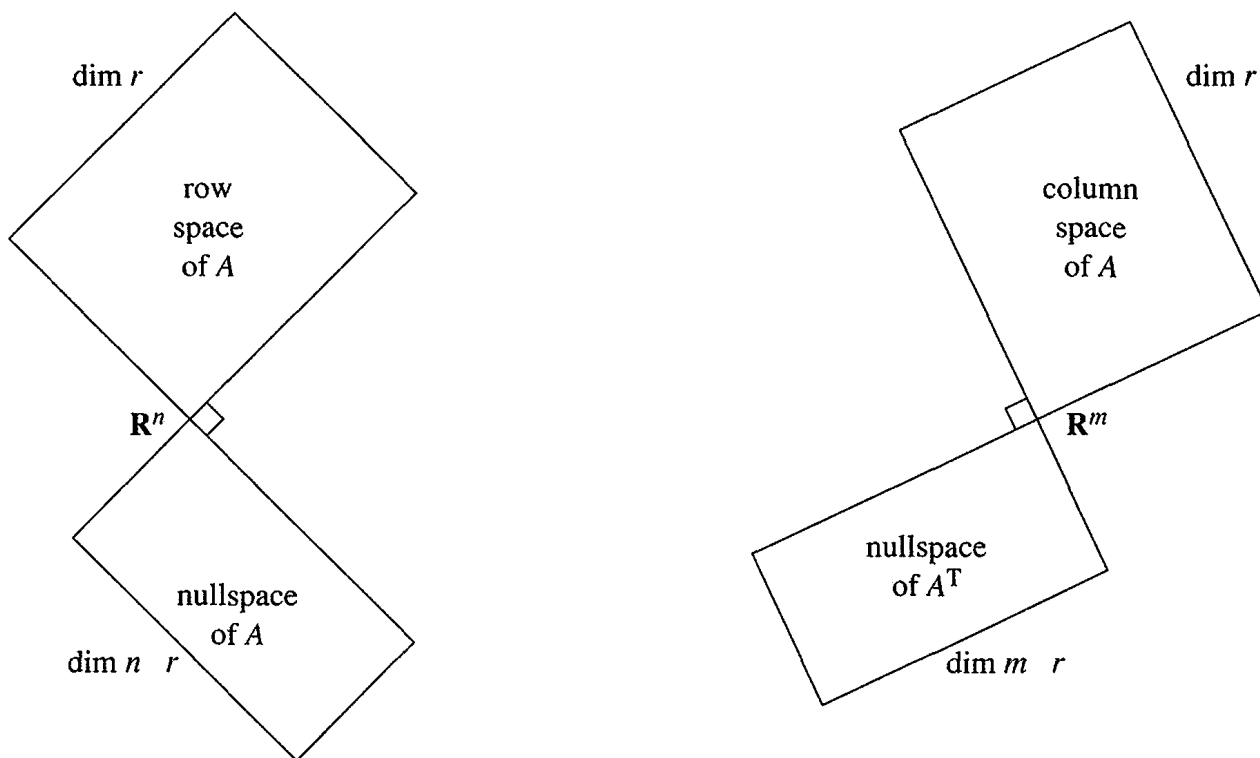


Figure 8.3: **Big picture:** The four subspaces with their dimensions and orthogonality.

Start with an m by n matrix. Its columns are vectors in $\mathbf{R}^m$. Their linear combinations produce the **column space** $C(A)$, a subspace of $\mathbf{R}^m$. Those combinations are exactly the matrix-vector products Ax .

The rows of A are vectors in $\mathbf{R}^n$ (or they would be, if they were column vectors). Their linear combinations produce the **row space**. To avoid any inconvenience with rows, we transpose the matrix. The row space becomes $C(A^T)$, the column space of A^T .

The central questions of linear algebra come from these two ways of looking at the same numbers, by columns and by rows.

The **nullspace** $N(A)$ contains every x that satisfies $Ax = 0$ —this is a subspace of $\mathbf{R}^n$. The “**left**” **nullspace** contains all solutions to $A^T y = 0$. Now y has m components, and $N(A^T)$ is a subspace of $\mathbf{R}^m$. Written as $y^T A = 0^T$, we are combining rows of A to produce the zero row. The four subspaces are illustrated by Figure 8.3, which shows $\mathbf{R}^n$ on one side and $\mathbf{R}^m$ on the other. The link between them is A .

The information in that figure is crucial. First come the dimensions, which obey the two central laws of linear algebra:

$$\dim C(A) = \dim C(A^T) \quad \text{and} \quad \dim C(A) + \dim N(A) = n.$$

When the row space has dimension r , the nullspace has dimension $n - r$. Elimination leaves these two spaces unchanged, and the echelon form U gives the dimension count. There are r rows and columns with pivots. There are $n - r$ free columns without pivots, and those lead to vectors in the nullspace.

This review of the subspaces applies to any matrix A —only the example was special. Now we concentrate on that example. It is the incidence matrix for a particular graph, and we look to the graph for the meaning of every subspace.

Directed Graphs and Incidence Matrices

Figure 8.4 displays a *graph* with $m = 6$ edges and $n = 4$ nodes, so the matrix A is 6 by 4. It tells which nodes are connected by which edges. The entries -1 and $+1$ also tell the direction of each arrow (this is a *directed* graph). The first row $-1, 1, 0, 0$ of A gives a record of the first edge from node 1 to node 2:

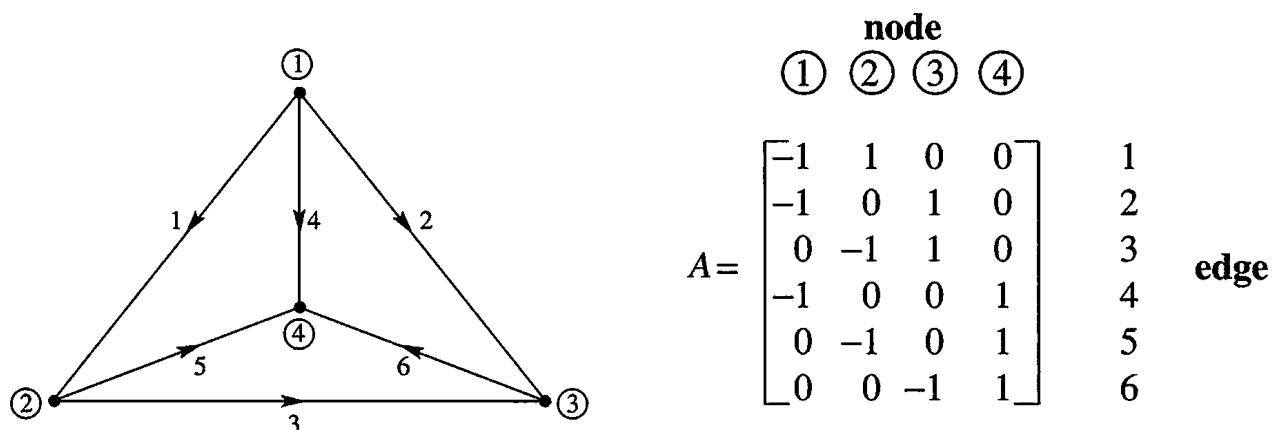


Figure 8.4a: Complete graph with $m = 6$ edges and $n = 4$ nodes.

Row numbers are edge numbers, column numbers are node numbers.

You can write down A immediately by looking at the graph.

The second graph has the same four nodes but only three edges. Its incidence matrix is 3 by 4:

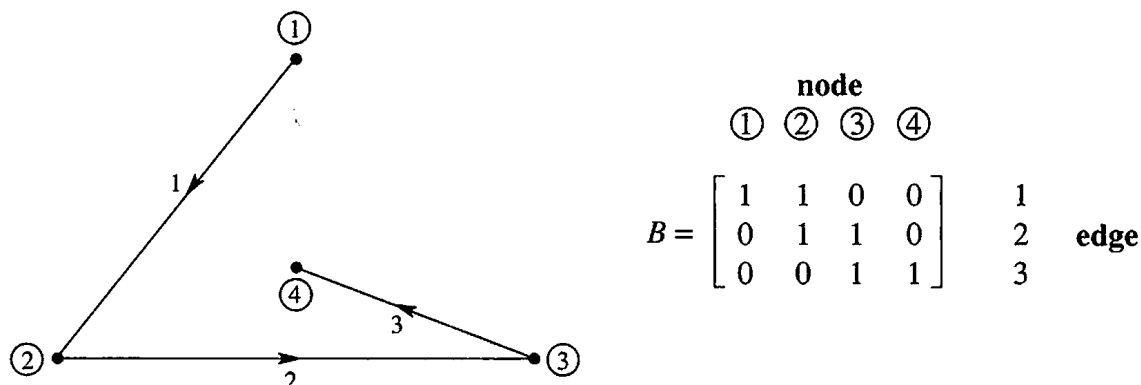


Figure 8.4b: Tree with 3 edges and 4 nodes and no loops.

The first graph is *complete*—every pair of nodes is connected by an edge. The second graph is a *tree*—the graph has *no closed loops*. Those graphs are the two extremes, the maximum number of edges is $\frac{1}{2}n(n-1)$ and the minimum (a tree) is $m = n - 1$.

The rows of B match the nonzero rows of U —the echelon form found earlier. **Elimination reduces every graph to a tree.** The loops produce zero rows in U . Look at the loop from edges 1, 2, 3 in the first graph, which leads to a zero row:

$$\begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Those steps are typical. When two edges share a node, elimination produces the “shortcut edge” without that node. If the graph already has this shortcut edge, elimination gives a row of zeros. When the dust clears we have a tree.

An idea suggests itself: **Rows are dependent when edges form a loop.** Independent rows come from trees. This is the key to the row space. We are assuming that the graph is connected, and it makes no fundamental difference which way the arrows go. On each edge, flow with the arrow is “positive.” Flow in the opposite direction counts as negative. The flow might be a current or a signal or a force—or even oil or gas or water.

For the column space we look at Ax , which is a vector of differences:

$$Ax = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_1 \\ x_3 - x_2 \\ x_4 - x_1 \\ x_4 - x_2 \\ x_4 - x_3 \end{bmatrix}. \quad (1)$$

The unknowns x_1, x_2, x_3, x_4 represent **potentials** or **voltages** at the nodes. Then Ax gives the **potential differences** or **voltage differences** across the edges. It is these differences that cause flows. We now examine the meaning of each subspace.

1 The nullspace contains the solutions to $Ax = 0$. All six potential differences are zero. This means: *All four potentials are equal.* Every x in the nullspace is a constant vector (c, c, c, c) . The nullspace of A is a line in $\mathbf{R}^n$ —its dimension is $n - r = 1$.

The second incidence matrix B has the same nullspace. It contains $(1, 1, 1, 1)$:

$$Bx = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

We can raise or lower all potentials by the same amount c , without changing the differences. There is an “arbitrary constant” in the potentials. Compare this with the same statement for functions. We can raise or lower $f(x)$ by the same amount C , without changing its derivative. There is an arbitrary constant C in the integral.

Calculus adds “ $+C$ ” to indefinite integrals. Graph theory adds (c, c, c, c) to the vector x of potentials. Linear algebra adds any vector x_n in the nullspace to one particular solution of $Ax = b$.

The “+C” disappears in calculus when the integral starts at a known point $x = a$. Similarly the nullspace disappears when we set $x_4 = 0$. The unknown x_4 is removed and so are the fourth columns of A and B . Electrical engineers would say that node 4 has been “grounded.”

2 The *row space* contains all combinations of the six rows. Its dimension is certainly not six. The equation $r + (n - r) = n$ must be $3 + 1 = 4$. The rank is $r = 3$, as we also saw from elimination. After 3 edges, we start forming loops! The new rows are not independent.

How can we tell if $v = (v_1, v_2, v_3, v_4)$ is in the row space? The slow way is to combine rows. The quick way is by orthogonality:

v is in the row space if and only if it is perpendicular to $(1, 1, 1, 1)$ in the nullspace.

The vector $v = (0, 1, 2, 3)$ fails this test—its components add to 6. The vector $(-6, 1, 2, 3)$ passes the test. It lies in the row space because its components add to zero. It equals $6(\text{row } 1) + 5(\text{row } 3) + 3(\text{row } 6)$.

Each row of A adds to zero. This must be true for every vector in the row space.

3 The *column space* contains all combinations of the four columns. We expect three independent columns, since there were three independent rows. The first three columns are independent (so are any three). But the four columns add to the zero vector, which says again that $(1, 1, 1, 1)$ is in the nullspace. *How can we tell if a particular vector b is in the column space of an incidence matrix?*

First answer Try to solve $Ax = b$. That misses all the insight. As before, orthogonality gives a better answer. We are now coming to Kirchhoff’s two famous laws of circuit theory—the voltage law and current law. Those are natural expressions of “laws” of linear algebra. It is especially pleasant to see the key role of the left nullspace.

Second answer Ax is the vector of differences in equation (1). If we add differences around a closed loop in the graph, the cancellation leaves zero. Around the big triangle formed by edges 1, 3, -2 (the arrow goes backward on edge 2) the differences cancel:

$$\text{Voltage Law} \quad (x_2 - x_1) + (x_3 - x_2) - (x_3 - x_1) = 0.$$

The components of Ax add to zero around every loop. When b is in the column space of A , it must obey the same law:

$$\text{Kirchhoff's Law:} \quad b_1 + b_3 - b_2 = 0.$$

By testing each loop, we decide whether b is in the column space. $Ax = b$ can be solved exactly when the components of b satisfy all the same dependencies as the rows of A . Then elimination leads to $0 = 0$, and $Ax = b$ is consistent.

4 The *left nullspace* contains the solutions to $A^T y = 0$. Its dimension is $m - r = 6 - 3$:

$$\begin{array}{l} \text{Current} \\ \text{Law (KCL)} \end{array} \quad A^T y = \begin{bmatrix} -1 & -1 & 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \quad (2)$$

The true number of equations is $r = 3$ and not $n = 4$. Reason: The four equations add to $0 = 0$. The fourth equation follows automatically from the first three.

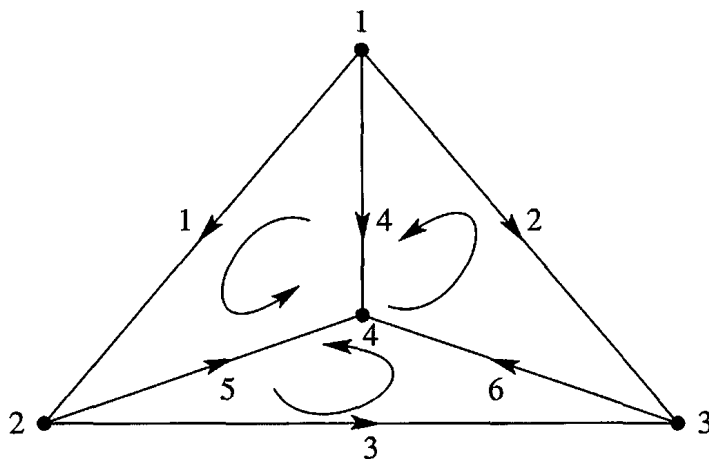
What do the equations mean? The first equation says that $-y_1 - y_2 - y_4 = 0$. *The net flow into node 1 is zero.* The fourth equation says that $y_4 + y_5 + y_6 = 0$. *Flow into the node minus flow out is zero.* The equations $A^T y = 0$ are famous and fundamental:

Kirchhoff's Current Law: *Flow in equals flow out at each node.*

This law deserves first place among the equations of applied mathematics. It expresses “conservation” and “continuity” and “balance.” Nothing is lost, nothing is gained. When currents or forces are in equilibrium, the equation to solve is $A^T y = 0$. Notice the beautiful fact that the matrix in this balance equation is the transpose of the incidence matrix A .

What are the actual solutions to $A^T y = 0$? The currents must balance themselves. The easiest way is to **flow around a loop**. If a unit of current goes around the big triangle (forward on edge 1, forward on 3, backward on 2), the vector is $y = (1, -1, 1, 0, 0, 0)$. This satisfies $A^T y = 0$. *Every loop current is a solution to the Current Law.* Around the loop, flow in equals flow out at every node. A smaller loop goes forward on edge 1, forward on 5, back on 4. Then $y = (1, 0, 0, -1, 1, 0)$ is also in the left nullspace.

We expect three independent y 's, since $6 - 3 = 3$. The three small loops in the graph are independent. The big triangle seems to give a fourth y , but it is the sum of flows around the small loops. The small loops give a basis for the left nullspace.



$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

small loops big loop

Summary The incidence matrix A comes from a connected graph with n nodes and m edges. The row space and column space have dimensions $n - 1$. The nullspaces of A and A^T have dimension 1 and $m - n + 1$:

- 1 The constant vectors $(c, c, \dots, c)$ make up the nullspace of A .
- 2 There are $r = n - 1$ independent rows, using edges from any tree.
- 3 **Voltage law:** The components of Ax add to zero around every loop.
- 4 **Current law:** $A^T y = 0$ is solved by loop currents. $N(A^T)$ has dimension $m - r$.
There are $m - r = m - n + 1$ independent loops in the graph.

For every graph in a plane, linear algebra yields *Euler's formula*:

$$(\text{number of nodes}) - (\text{number of edges}) + (\text{number of small loops}) = 1.$$

This is $n - m + (m - n + 1) = 1$. The graph in our example has $4 - 6 + 3 = 1$.

A single triangle has (3 nodes) $-$ (3 edges) $+ (1 \text{ loop})$. On a 10-node tree with 9 edges and no loops, Euler's count is $10 - 9 + 0$. All planar graphs lead to the answer 1.

Networks and $A^T C A$

In a real network, the current y along an edge is the product of two numbers. One number is the difference between the potentials x at the ends of the edge. This difference is Ax and it drives the flow. The other number is the “*conductance*” c —which measures how easily flow gets through.

In physics and engineering, c is decided by the material. For electrical currents, c is high for metal and low for plastics. For a superconductor, c is nearly infinite. If we consider elastic stretching, c might be low for metal and higher for plastics. In economics, c measures the capacity of an edge or its cost.

To summarize, the graph is known from its “connectivity matrix” A . This tells the connections between nodes and edges. A *network* goes further, and assigns a conductance c to each edge. These numbers $c_1, \dots, c_m$ go into the “conductance matrix” C —which is diagonal.

For a network of resistors, the conductance is $c = 1/(\text{resistance})$. In addition to Kirchhoff's Laws for the whole system of currents, we have Ohm's Law for each particular current. Ohm's Law connects the current y_1 on edge 1 to the potential difference $x_2 - x_1$ between the nodes:

Ohm's Law: *Current along edge = conductance times potential difference.*

Ohm's Law for all m currents is $y = -CAx$. The vector Ax gives the potential differences, and C multiplies by the conductances. Combining Ohm's Law with Kirchhoff's Current

Law $A^T y = 0$, we get $A^T C A x = 0$. This is *almost* the central equation for network flows. The only thing wrong is the zero on the right side! The network needs power from outside—a voltage source or a current source—to make something happen.

Note about signs In circuit theory we change from Ax to $-Ax$. The flow is from higher potential to lower potential. There is (positive) current from node 1 to node 2 when $x_1 - x_2$ is positive—whereas Ax was constructed to yield $x_2 - x_1$. The minus sign in physics and electrical engineering is a plus sign in mechanical engineering and economics. Ax versus $-Ax$ is a general headache but unavoidable.

Note about applied mathematics Every new application has its own form of Ohm's law. For elastic structures $y = CAx$ is Hooke's law. The stress y is (elasticity C) times (stretching Ax). For heat conduction, Ax is a temperature gradient. For oil flows it is a pressure gradient. There is a similar law in Section 8.6 for least squares regression in statistics.

My textbooks *Introduction to Applied Mathematics* and *Computational Science and Engineering* (Wellesley-Cambridge Press) are practically built on $A^T C A$. This is the key to equilibrium in matrix equations and also in differential equations. Applied mathematics is more organized than it looks. *I have learned to watch for $A^T C A$.*

We now give an example with a current source. Kirchhoff's Law changes from $A^T y = 0$ to $A^T y = f$, to balance the source f from outside. *Flow into each node still equals flow out.* Figure 8.5 shows the network with its conductances $c_1, \dots, c_6$, and it shows the current source going into node 1. The source comes out at node 4 to keep the balance (in = out). The problem is: **Find the currents $y_1, \dots, y_6$ on the six edges.**

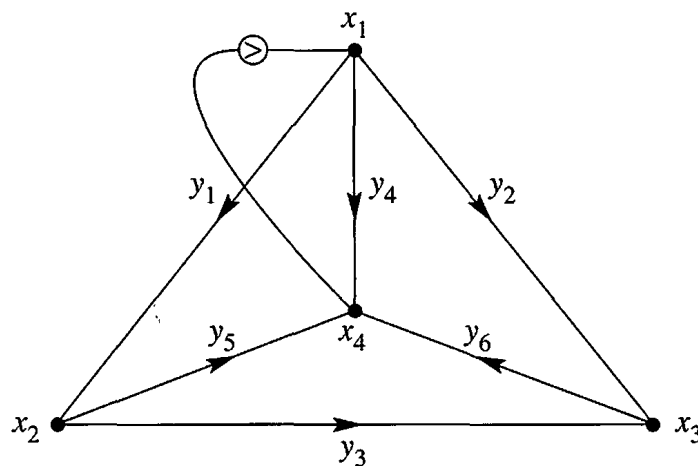


Figure 8.5: The currents in a network with a source S into node 1.

Example 1 All conductances are $c = 1$, so that $C = I$. A current y_4 travels directly from node 1 to node 4. Other current goes the long way from node 1 to node 2 to node 4 (this is $y_1 = y_5$). Current also goes from node 1 to node 3 to node 4 (this is $y_2 = y_6$). We can find the six currents by using special rules for symmetry, or we can do it right by using

$A^T C A$. Since $C = I$, this matrix is $A^T A$, the **graph Laplacian matrix**:

$$\begin{bmatrix} -1 & -1 & 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

That last matrix is not invertible! We cannot solve for all four potentials because $(1, 1, 1, 1)$ is in the nullspace. One node has to be grounded. Setting $x_4 = 0$ removes the fourth row and column, and this leaves a 3 by 3 invertible matrix. Now we solve $A^T C A x = f$ for the unknown potentials x_1, x_2, x_3 , with source S into node 1:

$$\text{Voltages} \quad \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} S \\ 0 \\ 0 \end{bmatrix} \quad \text{gives} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} S/2 \\ S/4 \\ S/4 \end{bmatrix}.$$

Ohm's Law $y = -C A x$ yields the six currents. Remember $C = I$ and $x_4 = 0$:

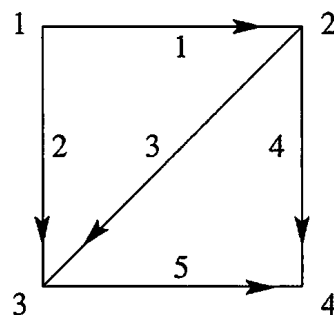
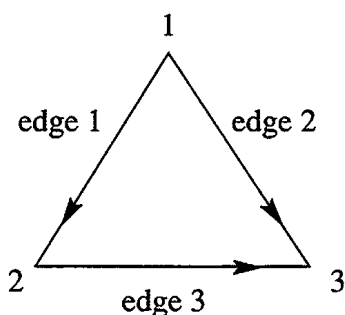
$$\text{Currents} \quad \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix} = - \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} S/2 \\ S/4 \\ S/4 \\ 0 \end{bmatrix} = \begin{bmatrix} S/4 \\ S/4 \\ 0 \\ S/2 \\ S/4 \\ S/4 \end{bmatrix}.$$

Half the current goes directly on edge 4. That is $y_4 = S/2$. No current crosses from node 2 to node 3. Symmetry indicated $y_3 = 0$ and now the solution proves it.

The same matrix $A^T A$ appears in least squares. Nature distributes the currents to minimize the heat loss. Statistics chooses $\hat{x}$ to minimize the least squares error.

Problem Set 8.2

Problems 1–7 and 8–14 are about the incidence matrices for these graphs.



- 1 Write down the 3 by 3 incidence matrix A for the triangle graph. The first row has -1 in column 1 and $+1$ in column 2. What vectors (x_1, x_2, x_3) are in its nullspace? How do you know that $(1, 0, 0)$ is not in its row space?
- 2 Write down A^T for the triangle graph. Find a vector y in its nullspace. The components of y are currents on the edges—how much current is going around the triangle?
- 3 Eliminate x_1 and x_2 from the third equation to find the echelon matrix U . What tree corresponds to the two nonzero rows of U ?

$$-x_1 + x_2 = b_1$$

$$-x_1 + x_3 = b_2$$

$$-x_2 + x_3 = b_3.$$

- 4 Choose a vector (b_1, b_2, b_3) for which $Ax = b$ can be solved, and another vector b that allows no solution. How are those b 's related to $y = (1, -1, 1)$?
- 5 Choose a vector (f_1, f_2, f_3) for which $A^T y = f$ can be solved, and a vector f that allows no solution. How are those f 's related to $x = (1, 1, 1)$? The equation $A^T y = f$ is Kirchhoff's _____ law.
- 6 Multiply matrices to find $A^T A$. Choose a vector f for which $A^T A x = f$ can be solved, and solve for x . Put those potentials x and the currents $y = -Ax$ and current sources f onto the triangle graph. Conductances are 1 because $C = I$.
- 7 With conductances $c_1 = 1$ and $c_2 = c_3 = 2$, multiply matrices to find $A^T C A$. For $f = (1, 0, -1)$ find a solution to $A^T C A x = f$. Write the potentials x and currents $y = -C A x$ on the triangle graph, when the current source f goes into node 1 and out from node 3.
- 8 Write down the 5 by 4 incidence matrix A for the square graph with two loops. Find one solution to $Ax = 0$ and two solutions to $A^T y = 0$.
- 9 Find two requirements on the b 's for the five differences $x_2 - x_1, x_3 - x_1, x_3 - x_2, x_4 - x_2, x_4 - x_3$ to equal b_1, b_2, b_3, b_4, b_5 . You have found Kirchhoff's _____ law around the two _____ in the graph.
- 10 Reduce A to its echelon form U . The three nonzero rows give the incidence matrix for what graph? You found one tree in the square graph—find the other seven trees.
- 11 Multiply matrices to find $A^T A$ and guess how its entries come from the graph:
 - (a) The diagonal of $A^T A$ tells how many _____ into each node.
 - (b) The off-diagonals -1 or 0 tell which pairs of nodes are _____.
- 12 Why is each statement true about $A^T A$? Answer for $A^T A$ not A .
 - (a) Its nullspace contains $(1, 1, 1, 1)$. Its rank is $n - 1$.